Energy Research & Social Science

Socioeconomic determinants of cooking fuel choice: evidence from the path toward energy sustainability in a developing country

--Manuscript Draft--

Manuscript Number:	ERSS-D-25-01928
Article Type:	Review Article
Section/Category:	Social determinants and demographics
Keywords:	Socioeconomic factors; Cooking fuel; Renewable energy; Sustainability; Developing country
Corresponding Author:	Brayan Tillaguango Holy Spirit University of Specialties Loja, ECUADOR
First Author:	Alex Uyaguari
Order of Authors:	Alex Uyaguari
	Brayan Tillaguango
	Rafael Alvarado
	Pavan Kumar
	Munir Ahmad
Abstract:	In developing countries, the persistent use of polluting fuels such as firewood and charcoal for cooking represents a serious environmental challenge with consequences for public health. In Ecuador, only 1.4% of the population uses renewable sources for cooking, while subsidies for imported liquefied petroleum gas continue to limit the energy transition. The objective of this research is to examine the effect of household socioeconomic conditions on cooking fuel choice using nationally representative microdata and multinomial logit models. The results show that, in rural areas, factors such as educational level, ethnic origin, number of rooms, and access to basic services increase the likelihood of using liquefied petroleum gas, while older age and marital status other than married reduce it. In urban areas, the likelihood of using electricity is positively associated with being male, having a higher education, being white, owning a home, and having internet access. These relationships remain robust when using ordered logit and generalized ordered logit models, which show asymmetric effects on the partial proportional odds of cooking fuel use. Overall, the findings suggest that public policies should focus on improving access to basic services in rural areas and strengthening infrastructure and human capital formation in urban areas to advance toward a sustainable energy transition, in line with Sustainable Development Goal 7.
Suggested Reviewers:	Santiago Ochoa-Moreno, PhD in economics University professor, Private Technical University of Loja wsochoa@utpl.edu.ec He has several publications on social and environmental issues.
	Stefania Pinzon, Magister Researcher, Holy Spirit University of Specialties yajairapinzon61@gmail.com Researcher in environmental and social fields
	Pablo Ponce, Magister University professor, National University of Loja pablo.ponce@unl.edu.ec Ambinetal and social thermal researcher
	Lizeth Cuesta, Economist Researcher, National University of Loja gisselle.cuesta@unl.edu.ec Researcher in environmental and social issues
	Diego Ochoa-Moreno, PhD in economics

Powered by Editorial Manager® and ProduXion Manager® from Aries Systems Corporation

	University professor, Private Technical University of Loja daochoa@utpl.edu.ec Researcher in environmental and social issues
	Zahoor Ahmed, PhD in economics University professor, Balochistan University of Engineering & Technology Khuzdar zahoorahmed83@yahoo.com Researcher in environmental and social issues
Opposed Reviewers:	

Powered by Editorial Manager® and ProduXion Manager® from Aries Systems Corporation

To:
The Editor
Energy Research & Social Science
Subject: Paper Submission

Dear Professor,

In the attached file, we submitted the paper entitled "**Socioeconomic determinants of cooking fuel choice: evidence from the path toward energy sustainability in a developing country**" for review process at **Energy Research & Social Science.** We confirm that we have not submitted this manuscript for possible publication anywhere else. It would be pleasure to have decision soon.

Thanks.
Sincerely,
The Authors

Corresponding author:

Brayan Tillaguango
Esai Business School, Universidad Espíritu Santo,
btillaguango@uees.edu.ec

The data are Available at:

https://www.ecuadorencifras.gob.ec/encuesta-nacional-multiproposito-de-hogares-2018/

Abstract

In developing countries, the persistent use of polluting fuels such as firewood and charcoal for cooking represents a serious environmental challenge with consequences for public health. In Ecuador, only 1.4% of the population uses renewable sources for cooking, while subsidies for imported liquefied petroleum gas continue to limit the energy transition. The objective of this research is to examine the effect of household socioeconomic conditions on cooking fuel choice using nationally representative microdata and multinomial logit models. The results show that, in rural areas, factors such as educational level, ethnic origin, number of rooms, and access to basic services increase the likelihood of using liquefied petroleum gas, while older age and marital status other than married reduce it. In urban areas, the likelihood of using electricity is positively associated with being male, having a higher education, being white, owning a home, and having internet access. These relationships remain robust when using ordered logit and generalized ordered logit models, which show asymmetric effects on the partial proportional odds of cooking fuel use. Overall, the findings suggest that public policies should focus on improving access to basic services in rural areas and strengthening infrastructure and human capital formation in urban areas to advance toward a sustainable energy transition, in line with Sustainable Development Goal 7.

Manuscript (without Author Details)

Socioeconomic determinants of cooking fuel choice: evidence from the path toward energy sustainability in a developing country

Alex Uyaguari (alex.uyaguari@unl.edu.ec)
Universidad Nacional de Loja, Loja 110150, Ecuador

Brayan Tillaguango (btillaguango@uees.edu.ec)
Esai Business School, Universidad Espíritu Santo, Samborondon 091650, Ecuador.

Rafael Alvarado (Corresponding autor) (jose.r.alvarado@unl.edu.ec)
Universidad Nacional de Loja, Loja 110150, Ecuador

Pavan Kumar (pvnkmr625@gmail.com)
College of Community Science, Central Agricultural University, Meghalaya, Tura, 794005, India

Munir Ahmad (munirahmad@nbufe.edu.cn)
Ningbo University of Finance and Economics, Ningbo 315175, Zhejiang, China

Socioeconomic determinants of cooking fuel choice: evidence from the path toward energy sustainability in a developing country

Abstract

In developing countries, the persistent use of polluting fuels such as firewood and charcoal for cooking represents a serious environmental challenge with consequences for public health. In Ecuador, only 1.4% of the population uses renewable sources for cooking, while subsidies for imported liquefied petroleum gas continue to limit the energy transition. The objective of this research is to examine the effect of household socioeconomic conditions on cooking fuel choice using nationally representative microdata and multinomial logit models. The results show that, in rural areas, factors such as educational level, ethnic origin, number of rooms, and access to basic services increase the likelihood of using liquefied petroleum gas, while older age and marital status other than married reduce it. In urban areas, the likelihood of using electricity is positively associated with being male, having a higher education, being white, owning a home, and having internet access. These relationships remain robust when using ordered logit and generalized ordered logit models, which show asymmetric effects on the partial proportional odds of cooking fuel use. Overall, the findings suggest that public policies should focus on improving access to basic services in rural areas and strengthening infrastructure and human capital formation in urban areas to advance toward a sustainable energy transition, in line with Sustainable Development Goal 7.

Keywords: Socioeconomic factors; Cooking fuel; Renewable energy; Sustainability; Developing country.

1. Introduction

The type of fuel used for cooking in homes directly influences environmental quality, public health, and household finances. Several studies have identified income levels and human capital as key factors in the adoption of clean cooking fuels (Dongzagla and Adams, 2022; Oyeniran and Isola, 2023; Qiu et al., 2023). The use of solid fuels, such as firewood or charcoal, contributes significantly to deforestation and environmental degradation. Meanwhile, the widespread use of liquefied petroleum gas (LPG), although less harmful locally, emits greenhouse gases that exacerbate the global climate crisis. These challenges underscore the urgent need for an energy transition that prioritizes cleaner fuels, such as electricity generated through induction cooktops. According to the World Health Organization (WHO, 2023), approximately 2.8 billion people worldwide continue to use polluting fuels such as firewood, coal, and kerosene, resulting in 3.2 million premature deaths in 2023, 7.4% of which were children under five years of age. In Ecuador, the use of polluting fuels for cooking continues to be a significant problem, especially in rural areas. According to the National Institute of Statistics and Census (INEC, 2018), 81.88% of rural households used LPG, 16.39% used firewood and coal, and only 1.43% used electricity. In urban areas, 95.59% used LPG, 2.92% used firewood and coal, and only 1.49% used electricity, reflecting the limited penetration of clean technologies, particularly in rural contexts. Furthermore, since 2018, a slight decrease in access to clean fuels has been observed. According to data from the World Bank (2020), 94.5% of the Ecuadorian population had access to these fuels in 2018, a figure that dropped to 94.45% in 2019 and 94.3% in 2020. At the global level, Stoner et al. (2021) warn that rural areas concentrate the highest use of polluting fuels, underscoring the need for differentiated policies that address territorial specificities in the transition to sustainable energy sources.

In order to promote the use of cleaner energy sources, Ecuador implemented the Efficient Cooking Program (PEC), which subsidized induction cooktops for approximately 600,000 families (Davi-Arderius et al., 2023). However, the results fell far short of the expectations of energy policymakers. Despite these efforts, LPG remains the primary source of cooking energy in the country due to its low cost and widespread availability. In rural areas, nearly 50% of households use firewood as their primary fuel, while in peri-urban areas the proportion reaches 20% (Gould et al., 2020). This continued use of traditional fuels is attributed to factors such as limited infrastructure for electric stoves, frequent equipment failures, concerns about the cost of electricity, and the continued existence of the LPG subsidy (Gould et al., 2018). Although this subsidy is justified for social purposes, it represents a significant burden for the State, given the country's dependence on LPG imports. According to ECLAC (2020), the average price of a 15-kg cylinder in Ecuador was USD 1.60, compared to a regional average of USD 13.31. Residential consumption represents 91% of total LPG demand, demonstrating its centrality in domestic cooking. This situation poses economic, environmental, and energy sustainability challenges. Therefore, it is essential to analyze the socioeconomic factors that influence the choice of cooking fuel, considering their implications for social equity and environmental quality.

In this context, the objective of this research is to analyze how socioeconomic factors influence household cooking fuel choices empirically. The dependent variable is categorical and consists of three main alternatives: LPG, firewood and charcoal (F&C), and electricity and other (E&O). Chi-square tests are used to identify significant differences between groups, as well as multinomial logit and generalized ordered logit (GOL) models to estimate changes in the probability of using each fuel based on household socioeconomic characteristics. This study contributes to the debate on residential energy transition in at least three dimensions. First, it explicitly addresses the heterogeneity between urban and rural areas, recognizing that household structural characteristics differ substantially between the two contexts. This distinction allows for more precise targeting of public policies toward specific groups, according to their particular conditions and constraints. Second, unlike much of the recent literature, which has focused its analyses on binary decisions or discrete choice models, this study considers multiple fuel alternatives, allowing for a more realistic approximation of household energy practices. To this end, multinomial logit and GOL models are estimated, along with their marginal effects, facilitating the identification of the mechanisms through which socioeconomic conditions influence fuel use decisions. Third, the characteristics of both the household head and the family environment are integrated into a single estimate, thus providing a more complete understanding of the determinants of energy choice at the household level.

2. Literature review

In recent years, academic interest in identifying the factors that determine cooking fuel choices has grown, allowing for advances in our understanding of the mechanisms underlying these energy decisions in households. Most studies are based on consumer behavior theory, which posits that households, as rational agents, tend to substitute traditional fuels for cleaner and more efficient options as their living conditions improve (Arowolo et al., 2018; Kapsalyamova et al., 2021). Along these lines, the energy ladder hypothesis argues that rising income drives a progressive transition toward modern energy sources, such as electricity, replacing more polluting ones (Hosier and Dowd, 1987; Van der Kroon et al., 2013; Gyamfi et al., 2020; Kumar, 2024; Barua et al., 2025). However, this hypothesis has been questioned, given that decisions about fuel use do not only respond to income, but also to social, cultural, institutional, and access factors, which reveals a more complex and multidimensional choice pattern (Masera et al., 2000; Alem et al., 2016; Ali et al., 2024; Bawakyillenuo et al., 2025).

Various studies agree that, in developing countries, the adoption of clean cooking fuels is strongly linked to improved household socioeconomic conditions. Factors such as increased income, educational level, housing tenure, and access to basic services are decisive in the transition to more modern and cleaner energy sources (Ma and Liao, 2018; Danlami et al., 2018; McLean et al., 2019; Ali and Khan, 2022; Murshed, 2023; Yunusa et al., 2024; Yao and Liu, 2025). During the COVID-19 lockdown, despite the drop in income, LPG use remained stable, while firewood use increased (Valarezo et al., 2023). Ishengoma and Igangula (2021) highlight that awareness of the risks of fuelwood use and access to information about the LPG market increase its adoption. However, Cabiyo et al. (2020) argue that households value LPG primarily for its time savings, rather than its health or environmental benefits. These findings reflect high heterogeneity in the determinants of household energy choices. Furthermore, recent empirical studies have documented the health risks associated with the use of polluting sources for cooking (Pantelic et al., 2023; Xu et al., 2023; Qiu et al., 2023; Amadu et al., 2023; Pu et al., 2023; Li et al., 2023; Liao et al., 2023; Fadly et al., 2023; Gunasegaran and Hasanuzzaman, 2025).

In parallel, various policymakers have promoted strategies to reduce the use of polluting fuels for cooking. Silva and Troncoso (2020) analyze the policies implemented in Bolivia, Ecuador, and El Salvador to facilitate access to clean energy, highlighting the role of economic incentives. In Ecuador, the transition to clean fuels has been supported by LPG subsidies. Gould (2023), using generalized additive mixed models, identifies that this transition contributed to reducing mortality from respiratory infections in children under five years of age. Meanwhile, Shupler et al. (2021) analyze the determinants of LPG supply and demand in peri-urban areas of Cameroon, Ghana, and Kenya, concluding that factors such as cylinder refilling, transportation costs, and the use of single-burner stoves better explain its use than socioeconomic factors. In Nigeria, Okereke et al. (2023) and Yunusa et al. (2024) highlight that the low use of LPG in rural areas is due to its cost, the need for accessories, and its limited availability. Membership in cooperative societies and participation in microfinance programs have also been shown to be important determinants of LPG adoption (Gill-Wiehl et al., 2021; Hsu et al., 2021; Nduka, 2021). These studies agree that LPG and firewood remain the primary cooking methods in households in developing countries.

Furthermore, some socioeconomic factors have been identified as significantly impacting households' decisions when choosing the type of fuel for cooking. For example, income level is one of the main determinants. According to Li et al. (2024), Zhang et al. (2023), and Zheng (2023), households with greater resources tend to use more sustainable fuels. The educational level, especially of women, is also crucial since education is associated with greater awareness of the risks of using traditional fuels on health and the environment (Bai et al., 2024; Huang et al., 2024; Aminu et al., 2024; Barik and Padhi, 2024). Access to energy infrastructure, such as electricity grids or LPG distribution points, facilitates the transition to cleaner energy sources. However, traditional fuels are more common in rural areas with limited infrastructure (Zheng, 2023). According to Yunusa et al. (2024), Zhang et al. (2023), and Li et al. (2024), household size also influences the fuel type decision, as larger households tend to opt for cheaper sources due to higher energy consumption. Geographic location also plays an important role: urban households, with better access to infrastructure, use more modern fuels than rural ones (Choudhury et al., 2024). The actions of families regarding the type of fuel used for cooking could have a spatial or temporal spillover, as various recent works have noted. For example, Gu (2022) found spatial dependence in the adoption of clean fuels, showing a significant influence on neighbors' adoption of such fuels in rural households. Onyeneke et al. (2023) use a distributed lag model and find that bidirectional causality exists between fuels, clean cooking technologies, rural population, GDP per capita, natural resource depletion, and damage from particle emissions. Likewise, Sreeja et al. (2023) examined the impact of access to clean cooking fuels on environmental degradation in BRICS nations. The authors use the Driscoll-Kraay (1998) method and found that energy use, trade openness, and urbanization promote greenhouse gas emissions. However, using clean cooking and foreign capital can help reduce environmental degradation in countries. Wang et al. (2024) found that crop residues are the primary fuel used in biomass burning, the choice of which is determined by meteorological factors. These results reflect that, despite the methodological heterogeneity of the research, the results indicate that the socioeconomic factors of individuals' homes influence the use of clean fuels for cooking.

Air quality improvements and limiting the negative impacts of polluting cooking methods require proper coordination among economic actors to maximize the benefits of the energy transition. Recent literature underscores the need for comprehensive and effective policies. Acheampong et al. (2023) highlight the role of globalization and governance in the adoption of clean technologies, while Liu et al. (2020) show that a centralized rural residence policy favors the use of clean fuels by increasing rural incomes and making traditional fuels more expensive. Notable policies include the photovoltaic poverty alleviation project in China, which, according to Liu et al. (2023), has promoted the use of more sustainable fuels. However, household preferences also play a role. In Peru, Nuño et al. (2023) report low acceptance of biomass stoves among users not involved in the program design. In India, Pelz et al. (2021) show that marginalized households face structural barriers to electricity access. Islam et al. (2023) warn that LPG subsidies can facilitate flexible payment schemes but also generate distortions that deepen inequality. This review reveals gaps in the literature, such as the scant consideration of multiple sources of cooking in the home and the limited attention to the link between cooking techniques and environmental problems in developing countries. The present study contributes by addressing these gaps through a methodology that allows estimating the probability of use of three traditional fuel types: charcoal, LPG, and electricity.

3. Statistical information and econometric strategy

3.1. Data sources and their characteristics

The dataset used in this study comes from the National Institute of Statistics and Censuses of Ecuador (INEC), specifically from the *Encuesta Multipropósito 2018*[1]. This statistical operation comprises seven databases, from which those relevant to the research objectives were selected: information on household members, education, household equipment, and housing and household characteristics. After a filtering process that consisted of keeping only records of household heads between the ages of 15 and 94, the final sample consisted of 25,028 households, representing both urban (15,175 households) and rural (9,853) areas. The survey provides a detailed understanding of the socioeconomic status of households and their members, which together provides a robust analytical framework for more accurately estimating the determinants of cooking fuel choices.

The dependent variable was constructed from the question addressed to the head of household: "In this household, do you primarily cook with gas?" The response options were "LPG," "Firewood or coal," and "Electricity." Due to this variable's categorical and multinomial nature, the use of econometric models that adequately capture its response structure was justified. Regarding the explanatory variables, two main blocks were considered. The first brings together the sociodemographic and economic characteristics of the head of household, defined as the person who makes the main decisions in household matters (Zhang & Hassen, 2017; Liao et al., 2019). In this regard, the impact of variables such as age, gender, marital status, ethnicity, educational level, and occupation of the head of household was analyzed, distinguishing between whether they work in the public or private sector. Likewise, household income was incorporated, which is in line with the previously discussed energy ladder hypothesis.

The second block comprises variables related to housing conditions, which reflect household lifestyles and needs. Previous evidence suggests that households with better housing conditions, such as access to clean water, modern sanitation, and electricity, tend to show greater health concerns and a more favorable disposition toward using clean cooking fuels (Sharma & Rahman, 2025). Based on data availability, variables such as home ownership, number of rooms, access to the internet, clean water, and electricity were included. To facilitate the interpretation of the quantitative variables, they were categorized. Monthly income was classified into five levels: low income (less than $90), lower-middle-income ($90.01–$142), middle income ($142.01–$203), upper-middle income ($203.01–$317), and high income (more than $317). Age was segmented into three groups: under 30, between 31 and 60, and over 60. Marital status was coded into two categories: married and other (which includes separated, divorced, widowed, in a common-law relationship, and single). The descriptive statistics for these variables are presented in Table 1.

Table-1: Selection of Variables and Descriptive Statistics

Variable	Description	Mean	S.D.	SK.	Kurtosis	Min.	Max.
Cooking fuel	0=F&C. 1= LGP. 2= E&O.	0.07 0.90 0.02	0.26 0.30 0.15	-1.24	9.78	0	2
Head of household							
Age	1= < 30 years. 2=31-60 years. 3= > 60 years.	0.10 0.60 0.30	0.30 0.49 0.46	-0.10	2.57	1	3

[1] The Multipurpose Survey is a national statistical operation that collects information on demographic, economic, and social aspects from a representative sample of 26,928 households distributed throughout Ecuador. Its main objective is to generate statistical data that allows monitoring the achievement of the objectives established in the National Development Plan (NDP) and other development agendas.

Marital status	0=Other case. 1=Married.	0.56 0.44	0.50 0.50	0.24	1.06	0	1
Gender	0=Women. 1=Man.	0.26 0.74	0.44 0.44	-1.11	4.43	0	1
Ethnicity	1=Indigenous. 2= Afro-Ecuadorian. 3= Black. 4=Mulato. 5=Montubio. 6=Mestizo. 7=White.	0.13 0.02 0.02 0.02 0.04 0.76 0.02	0.33 0.13 0.13 0.13 0.18 0.42 0.14	-1.78	4.43	1	7
Human capital	0= Illiterate. 1= Basic education. 2= Middle education. 3= Higher education.	0.06 0.47 0.33 0.14	0.23 0.50 0.47 0.35	0.30	2.41	0	3
Income	1= Low income (<$90). 2=Lower middle income ($90.01 - $142). 3= Middle income ($142.01 and $203). 3=Upper middle income ($203.01 and $317). 4= High income (>$317.01)	0.02 0.02 0.04 0.88	0.14 0.13 0.19 0.32	-3.48	14.82	1	4
Public sector employee	0= No. 1=Yes.	0.91 0.09	0.28 0.28	2.97	9.80	0	1
Housing conditions							
Homeowner.	0=No. 1=Yes.	0.16 0.84	0.37 0.37	-1.85	4.44	0	1
Number of rooms.	1= One. 2= Two. 3= Three. 4= Four. 5= >5.	0.26 0.37 0.26 0.08 0.03	0.44 0.48 0.44 0.28 0.16	0.60	2.87	1	5
Internet access.	0= No. 1=Yes.	0.68 0.32	0.47 0.47	0.79	1.62	0	1
Access to drinking water.	0= No. 1= Yes.	0.32 0.68	0.47 0.47	-0.76	1.58	0	1
Access to electricity.	0= No. 1=Yes.	0.02 0.98	0.14 0.14	-6.86	48.11	0	1
Source: Own elaboration with data from INEC (2018)							

Figure 1 shows the distribution of cooking fuel types in Ecuador, differentiating between rural and urban areas. In both areas, LPG predominates, with a prevalence of 81.88% in rural areas and 95.59% in urban areas. In rural areas, 16.39% of households still use unclean solid fuels, such as firewood or coal. Electricity use is low in both areas, although slightly higher in urban areas (2.49%) compared to rural areas (1.43%), which is consistent with the findings of Gould et al. (2020). These results challenge the energy ladder hypothesis, as they show that households do not necessarily move toward more modern sources, but rather opt for more accessible and affordable ones. In Ecuador, this preference for LPG may be influenced by its subsidy policy, underscoring the importance of considering social, economic, and structural factors in the analysis of household energy choices.

Figure-1: composition of fuels used for cooking in Ecuador (2018)

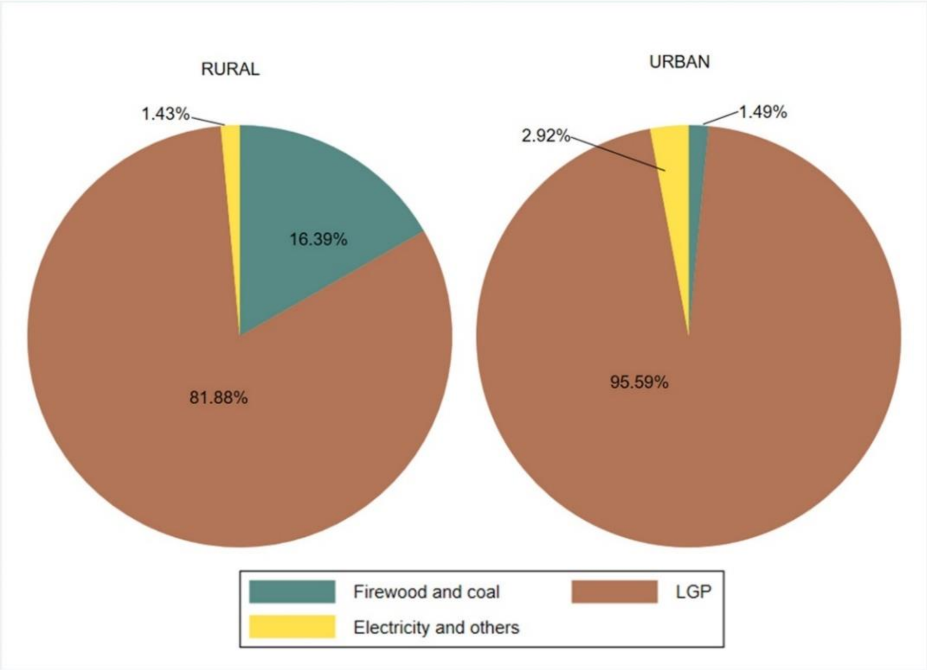

Table 2 reports the results of the test for collinearity between covariates using the Variance Inflation Factor (VIF). With an average value of 1.18, no severe multicollinearity is evident among the included variables. This absence is essential to ensure the efficiency of the estimators and properly isolate the effect of each covariate on fuel choice in Ecuadorian households.

Table-2: Collinearity Test

Variables	VIF	SQRT-VIF	Tolerance	R-Squared
Head of household				
Age	1.23	1.11	0.81	0.19
Marital status	1.31	1.14	0.76	0.24
Gender	1.23	1.11	0.81	0.19
Ethnicity	1.14	1.07	0.72	0.12
Human capital	1.39	1.18	0.97	0.28
Income	1.04	1.02	0.88	0.03
Public sector employee	1.13	1.06	0.88	0.12
Housing conditions				
Homeowner.	1.14	1.07	0.88	0.12
Number of rooms.	1.14	1.07	0.88	0.12
Internet access.	1.14	1.07	0.87	0.12
Access to drinking water.	1.22	1.11	0.81	0.18
Access to electricity.	1.07	1.03	0.94	0.06
Mean VIF	1.18			

3.2. Model specification

This study classifies cooking fuels into three groups: "Wood or coal," "LPG," and "Electricity or other." Since households often use multiple fuels simultaneously, a binary logit model is inappropriate. Although an ordered logit model could be considered when ranking fuels by cleanliness or efficiency, heterogeneous household preferences and the mixed use of energy sources make strict ranking difficult, potentially violating the parallel lines assumption, which is frequently breached in practice (Long and Freese, 2006). Therefore, a multinomial logit (MNL) model is employed, which allows greater flexibility when dealing with unordered categorical dependent variables (Greene, 2012). In this approach, probabilities are estimated relative to a base category, and results are interpreted based on the marginal effects.

$$Pr(Y_i = j|w_i) = P_{ij} = \frac{exp(w_i^{'} \propto_j)}{1 + \sum_{k=1}^{J} exp(w_i^{'} \propto_k)}, \quad j = 0,1,\ldots,J. \qquad (1)$$

In Equation 1 we estimate the probability that the element or individual Y_i belongs to category j based on the characteristics w_i. This probability results from the division of the exponential of the probability equation of belonging to a reference category j divided by 1 plus the sum of the conditional probabilities for all the categories. J denotes the base comparison category. If the dependent variable has J categories, J-1 vectors of parameter are estimated, denoted by $\propto_j$ and $\propto_k$. Considering that our dependent variable has three categories: (0) firewood and charcoal, (1) LPG, and (2) electricity, the firewood and charcoal category has been selected as the type of comparison fuel. The probability prediction equations are stated as follows:

$$P1_i = \frac{exp(w_i^{'} \propto_1)}{1 + exp(w_i^{'} \propto_1) + exp(w_i^{'} \propto_2)} \qquad (2).$$

$$P2_i = \frac{exp(w_i^{'} \propto_2)}{1 + exp(w_i^{'} \propto_1) + exp(w_i^{'} \propto_2)} \qquad (3).$$

$$P3_i = \frac{1}{1 + exp(w_i^{'} \propto_1) + exp(w_i^{'} \propto_2)} \qquad (4).$$

Given that there are three categories in the dependent variable, 3 vectors of parameters denoted as $\propto_1$ and $\propto_2$ are estimated, which are the effect of the explanatory variables on the log odds of belonging to the LPG reference category for Equation (2) and electricity and others for Equation (3) instead of the comparison category F&C (L&C). The reference category is in the numerator, while the denominator is the base category. Furthermore, $\propto_2$ encompasses all independent variables. Equations 2 and 3 present the models to be estimated as follows:

$$\begin{aligned} ln\left(\frac{P_{iGLP}}{P_{iL\&C}}\right) = & \propto_0^1 + \propto_1^1 Age_i + \propto_2^1 MS_i + \propto_3^1 Gender_i + \propto_4^1 Ethnicity + \propto_5^1 HC_i + \propto_6^1 IN_i \\ & + \propto_7^1 Empl_i + \propto_8^1 HO_i + \propto_9^1 Rooms_i + \propto_{10}^1 Inter_i + \propto_{11}^1 Water_i \\ & + \propto_{12}^1 Elec_i + e \end{aligned} \qquad (5)$$

$$\begin{aligned} ln\left(\frac{P_{iElectri}}{P_{iL\&C}}\right) = & \propto_0^1 + \propto_1^1 Age_i + \propto_2^1 MS_i + \propto_3^1 Gender_i + \propto_4^1 Ethnicity + \propto_5^1 HC_i \\ & + \propto_6^1 IN_i + \propto_7^1 Empl_i + \propto_8^1 HO_i + \propto_9^1 Rooms_i + \propto_{10}^1 Inter_i \\ & + \propto_{11}^1 Water_i + \propto_{12}^1 Elec_i + e \end{aligned} \qquad (6)$$

Equation 5 states that the logarithm of the probability of using LPG concerning firewood and charcoal is a function of the twelve explanatory variables (w_i times $\propto_i^1$). Equation 6 represents the logarithm of the probability of using electricity concerning F&C (w_i times $\propto_i^2$). The superscript 1 and 2 of the alpha parameter in the equations denote the alpha vector 1 and alpha vector 2, respectively. For interpretation, marginal effects will represent the change in the probability of belonging to a specific category due to a unit change in an explanatory variable. The advantage of this methodology is that it allows us to identify the factors that significantly affect the choice of the type of fuel for cooking. This fact makes it possible to formulate policies to reduce pollution and promote clean energy use in homes in developing countries.

4. Results and discussion

4.1. Socioeconomic characteristics of households and the result of chi-square Test

Table 2 presents the results of the chi-square (χ^2) test, which reveals significant differences between the explanatory variables and fuel choice by geographic area. The null hypothesis states no differences, while the alternative suggests the opposite. P values below 5% reject the null hypothesis. In rural areas, all variables except gender show a significant association with the fuel used. For example, age is significantly related to fuel type ($\chi^2 = 82,21$; $p < 0,05$): LPG is the most commonly used fuel across all age groups, especially among heads of household aged 31 to 60 (60% of LPG users). Electricity is uncommon among those under 30, while the use of firewood and coal is more common among adults. Regarding marital status, singles show greater use of clean fuels, such as LPG (53.33%) and electricity (58.04%), compared to married people. Ethnicity is also significantly associated with fuel choice ($\chi^2 = 115.03$; $p < 0.05$): mestizos predominate in the use of LPG (71.54%) and electricity (67.13%), while indigenous people show a considerably higher use of firewood and charcoal (63.09%). These findings are consistent with those reported by McLean et al. (2019) and Cabiyo et al. (2020), who highlight that indigenous communities tend to rely more on solid fuels.

Education is strongly associated with the use of F&C as cooking fuel ($\chi^2 = 328.21$; $p < 0.05$). 9.84% of household heads are illiterate, 60.54% have primary education, 24.12% have secondary education, and only 5.51% have higher education. Households with primary education account for the highest use of F&C (61.58%), followed by LPG (60.56%) and electricity (46.85%). Despite this, LPG is the most commonly used fuel at all education levels, while electricity is the least common. Even among those with higher education, the use of F&C exceeds that of electricity, with only 13% of electricity users belonging to this group. Regarding income, a significant, albeit weak, association with F&C use is observed ($\chi^2 = 43.65$; $p < 0.05$). LPG (84.49%) and electricity (86.01%) predominate in higher-income households. However, F&C use also increases with income, accounting for 89.62% of users. These results partially contrast with those of Onyeneke et al. (2023), who report a different relationship between income and fuel type in rural areas. However, they coincide with the findings of Ali and Khan (2022) and Wassie et al. (2021), except for the sex of the household head. Taken together, the results suggest that education and income, although relevant, are not exclusive determinants in cooking fuel choice. Recent studies highlight the role of social awareness as a key factor in households' sustainable energy decisions (Wang et al., 2023; Zhou and Ding, 2023; Severo et al., 2023).

Classical environmental economics argues that property rights are fundamental to protecting the environment (Pan et al., 2023). However, other factors also influence pollution. This research analyzes whether homeownership and public sector employment influence the use of sustainable cooking methods. Both variables show a significant but weak relationship with the use of F&C: 94.12% of household heads are not public employees and 94.22% are homeowners. In both groups, electricity use is low. In contrast, the number of rooms is strongly associated with F&C ($\chi^2 = 291.58$; $p < 0.05$), with lower use in households with more rooms, suggesting better structural conditions and higher income. These findings are consistent with Perros et al. (2022), who approached this issue from a behavioral perspective. Likewise, access to basic services such as the internet, water, and electricity is significantly related to fuel choice. In rural areas, only 11.46% of households have access to the internet, 42.82% to water, and 95.24% to electricity. LPG is concentrated in households with essential services, while F&C predominates in those without electricity access. These results are consistent with Pelz et al. (2021) and Liu et al. (2023), who identify energy inequalities in marginal rural areas. Along these lines, Onyeneke et al. (2023) recommend improving access to basic services as a condition for an effective energy transition.

In urban areas, age, marital status, and income did not show a significant association with F&C use ($p > 0.05$), while gender was marginally significant (at 10%). 71.49% of household heads were men, of whom 68.70% used F&C, 71.39% used LPG, and 76.22% used electricity. In contrast, variables such as ethnicity, educational level, public employment, home ownership, number of rooms, and access to basic services (internet, water, and electricity) were significantly associated with the type of fuel used. These results partially coincide with those of Wassie et al. (2021), Dongzagla and Adams (2022), and Ali and Khan (2022), who did identify a significant relationship between income and fuel type. In this urban sample, 10.43% of F&C users belonged to indigenous ethnic groups, and the majority had a secondary education. Furthermore, 94% of household heads report incomes above $317, and among them, electricity is the second most used fuel after LPG (94.44%). Employment is concentrated in the private sector (89%), and most households are homeowners (77.38%). As the number of rooms increases, the use of fuel oil and electricity as cooking energy sources decreases, with firewood being most common in two-room dwellings (36%). Access to basic services is higher in urban areas: internet (44.77%), water (84.08%), and electricity (99.33%), and is positively correlated with the use of electricity as a fuel. These findings are consistent with Danlami et al. (2018), who also highlight the role of socioeconomic and demographic factors in household fuel choice.

4.2. Factors affecting the households fuel choice

Table 4 presents the results of the Wald and likelihood ratio tests. The former assesses the individual significance of the coefficients of the explanatory variables. In rural households, gender, income, and home ownership are not significant; in urban areas, age, Afro-Ecuadorian and mulatto ethnicities, basic and secondary education, income, and dwellings with more than five rooms are not significant. For the total, income also shows no significance. This lack of effect could be attributed to factors such as basic infrastructure or cultural aspects, given that variables such as access to water, electricity, and ethnicity have high chi-square values. The significant variables consistently influence fuel choice. These results are consistent with previous studies (Gill-Wiehl et al., 2021; Ishengoma and Igangula, 2021; Qiu et al., 2023), which highlight the importance of sociodemographic factors and preferences in decisions about cooking methods. The likelihood ratio test assesses whether the elimination of a variable significantly affects the model's fit, comparing a full model with a restricted model. The null hypothesis (H_0) states that both models fit equally, while the alternative (H_1) indicates that the full model offers a better fit. High chi-square values suggest a significant difference. In rural areas, the most influential variables are age, marital status, ethnicity, education, and housing conditions (except property ownership). In urban areas, ethnicity, higher education, and access to electricity are not significant. These variables improve the model's predictive power. On the other hand, in rural areas, gender and income are not significant, and in urban areas, neither are age, the "Afro-Ecuadorian" and "mulatto" ethnicities, basic and secondary education, income, public employment, and housing with more than five rooms. Overall, income is not significant (p =

0.05). Factors such as age, marital status, gender, and education, along with access to water and electricity ($Chi^2 = 455.61$ and 285.84, respectively), internet access, number of rooms, and home ownership, are relevant. These findings partially coincide with previous studies (Gu, 2022; Li and Zhou, 2023). Table 5 presents the results of the Hausman test to assess compliance with the Independence of Irrelevant Alternatives (IIA) assumption in the multinomial logit model. The null hypothesis states that the relationship between choice options is constant and independent of the inclusion of other alternatives. High values of the Chi^2 statistic indicate a greater discrepancy with this premise. The results show that the F&C and LGP categories do not present significant evidence against the IIA assumption in any of the areas or the total population ($p > 0.05$), so the null hypothesis is not rejected. However, in the case of the E&O category for the total population, a violation of the assumption is detected ($p < 0.05$), which suggests dependence on the other alternatives (F&C and LGP).

Table-3: Distributions and Chi-Squared Test Statistics

Variables		Rural (n=9853)						Urban (n=15175)					
		F&C	LGP	E&O	N (%)	χ^2	P	F&C	LGP	E&O	N (%)	χ^2	P
Head of household													
Age	1	109 (6.61)	720 (8.93)	15 (10.56)	844 (9.57)	86.21	0.00	22 (9.65)	1528 (10.54)	45 (10.16)	1595(10.51)	7.62	0.11
	2	843 (51.15)	4875 (60.46)	82 (57.75)	5800 (58.87)			123 (53.95)	8890 (61.29)	274 (61.85)	9287 (61.20)		
	3	696 (42.23)	2468 (30.61)	45 (31.69)	3209 (32.57)			83 (36.40)	4086 (28.17)	124 (27.99)	4293 (28.29)		
Marital status	0	798 (47.90)	4360 (53.33)	83 (58.04)	5241 (52.49)	18.18	0.00	144 (62.61)	8531 (57.92)	268 (59.56)	9843 (58.04)	2.48	0.29
	1	868 (52.10)	3815 (46.67)	60 (41.96)	60 (41.96)			86 (37.39)	6198 (42.08)	182 (40.44)	6466 (41.96)		
Gender	0	336 (20.17)	1797 (21.98)	26 (18.18)	2159 (21.62)	3.70	0.16	72 (31.30)	4214 (28.61)	107 (23.78)	4393 (28.51)	5.90	0.05
	1	1330 (79.83)	6378 (78.02)	117 (81.82)	7825 (78.38)			158 (68.70)	10515 (71.39)	343 (76.22)	11016 (71.49)		
Ethnicity	1	1051 (63.09)	1463 (17.90)	29 (20.28)	2543 (25.47)	115.03	0.00	24 (10.43)	650 (4.41)	9 (2.00)	683 (4.43)	47.22	0.00
	2	7 (0.42)	87 (1.06)	1 (0.70)	95 (0.95)			11 (4.78)	300 (2.04)	12 (2.67)	323 (2.10)		
	3	6 (0.36)	109 (1.33)	3 (2.10)	118 (1.18)			5 (2.17)	328 (2.23)	11 (2.44)	344 (2.23)		
	4	6 (0.36)	104 (1.27)	1 (0.70)	111 (1.11)			8 (3.48)	331 (2.25)	6 (1.33)	345 (2.24)		
	5	40 (2.40)	427 (5.22)	12 (8.39)	479 (4.80)			4 (1.74)	406 (2.76)	14 (3.11)	424 (2.75)		
	6	544 (32.65)	5848 (71.54)	96 (67.13)	6488 (64.98)			173 (75.22)	12379 (84.05)	379 (84.22)	12931 (83.92)		
	7	12 (0.72)	137 (1.68)	1 (0.70)	150 (1.50)			5 (2.17)	335 (2.27)	19 (4.22)	359 (2.33)		
Human capital	0	333 (19.99)	635 (7.77)	14 (9.79)	982 (9.84)	328.21	0.00	16 (6.96)	475 (3.22)	10 (2.22)	501 (3.25)	95.10	0.00
	1	1026 (61.58)	4951 (60.56)	67 (46.85)	6044 (60.54)			107 (46.52)	5661 (38.43)	116 (25.78)	5884 (38.19)		
	2	280 (16.81)	2085 (25.50)	43 (30.07)	2408 (24.12)			80 (34.78)	5670 (38.50)	165 (36.67)	5915 (38.39)		
	3	27 (1.62)	504 (6.17)	19 (13.29)	550 (5.51)			27 (11.74)	2923 (19.85)	159 (35.33)	3109 (20.18)		
Income	1	42 (2.52)	202 (2.47)	6 (4.20)	250 (2.50)	43.65	0.00	4 (1.74)	231 (1.57)	6 (1.33)	241(1.56)	11.80	0.16
	2	36 (2.16)	191 (2.34)	4 (2.80)	231 (2.31)			4 (1.74)	202 (1.37)	4 (0.89)	210(1.36)		
	3	47 (2.82)	398 (4.87)	6 (4.20)	451 (4.52)			8 (3.48)	448 (3.04)	5 (1.11)	461(2.99)		
	4	48 (2.88)	477 (5.83)	4 (2.80)	529 (5.30)			5 (2.17)	550 (3.73)	10 (2.22)	565(3.67)		
	5	1493 (89.62)	6907 (84.49)	123 (86.01)	8523 (85.37)			209 (90.87)	13298 (90.28)	425 (94.44)	13932(90.41)		
Public sector employee	0	1629 (97.78)	7643 (93.49)	125 (87.41)	9397 (94.12)	57.76	0.00	210 (91.30)	13229 (89.82)	388 (86.22)	13827 (89.73)	6.75	0.03
	1	37 (2.22)	532 (6.51)	18 (12.59)	587 (5.88)			20 (8.70)	1500 (10.18)	62 (13.78)	1582 (10.27)		
Housing conditions													
Homeowner.	0	43 (2.58)	524 (6.41)	10 (6.99)	577(5,78)	37.65	0.00	61 (26.52)	3354 (22.77)	71 (15.78)	3486 (22.62)	14.23	0.00
	1	1623 (97.42)	7651 (93.59)	133 (93.01)	9407(94.22)			169 (73.48)	11375 (77.23)	379 (84.22)	11923 (77.38)		
Number of rooms.	1	761 (45.68)	2087 (25.53)	42 (29.37)	2890 (28.95)	291.58	0.00	95 (41.30)	3514 (23.86)	133 (29.56)	3742 (24.28)	55.22	0.00
	2	558 (33.49)	3300 (40.37)	56 (39.16)	3914 (39.20)			72 (31.30)	5379 (36.52)	126 (28.00)	5557 (36.19)		
	3	224 (13.45)	1863 (22.79)	34 (23.78)	2121 (21.24)			42 (18.26)	4191 (28.45)	134 (29.78)	4367 (28.34)		
	4	100 (/6.00)	687 (8.40)	8 (5.59)	795 (7.96)			14 (6.09)	1245 (8.45)	44 (9.78)	1303 (8.46)		
	5	23 (1.38)	238 (2.91)	3 (2.0)	264 (2.64)			7 (3.04)	400 (2.72)	13 (2.89)	420 (2.73)		
Internet access	0	1614 (96.88)	7111 (86.98)	115 (80.42)	8840 (88.54)	142.97	0.00	158 (68.70)	8153 (55.35)	200 (44.44)	8511 (55.23)	38.13	0.00
	1	52 (3.12)	1064 (13.02)	28 (19.58)	1144 (11.46)			72 (31.30)	6576 (44.65)	250 (55.56)	6898 (44.77)		

Access to drinking water.	0	1415 (84.93)	4223 (51.66)	71 (49.65)	5709 (57.18)	629.27	0.00	76 (33.04)	2315 (15.72)	62 (13.78)	2453 (15.92)	52.38	0.00
	1	251 (15.07)	3952 (48.34)	72 (50.35)	4275 (42.82)			154 (66.96)	12414 (84.28)	388 (86.22)	12956 (84.08)		
Access to electricity.	0	314 (18.85)	153 (1.87)	8 (5.59)	475 (4.76)	880.41	0.00	10 (4.35)	19 (0.13)	3 (0.67)	32 (0.21)	199.21	0.00
	1	1352 (81.15)	8022 (98.13)	135 (94.41)	9509 (95.24)			220 (95.65)	14710 (99.87)	447 (99.33)	15377 (99.79)		
		Total (n=25028)											
		F&C	LGP	E&O	N (%)	χ^2	P						
Head of household													
Age	1	131 (6.98)	2248 (9,96)	60 (10.26)	2439 (9.75)	132.18	0.00						
	2	966 (51.49)	13765 (61.00)	356 (60.85)	15087 (60.28)								
	3	779 (41.42)	6554 (29.04)	169 (28.89)	7502 (29.97)								
Marital status	0	942 (49.68)	12891 (56.28)	351 (59.19)	14184 (55.86)	33.66	0.00						
	1	954 (50.32)	10013 (43.72)	242 (40.81)	11209 (44.14)								
Gender	0	408 (21.52)	6011 (26.24)	133 (22.43)	6552 (25.80)	24.03	0.00						
	1	1448 (78.48)	16893 (73.76)	460 (77.57)	18641 (74.20)								
Ethnicity	1	1075 (56.70)	2113 (9.23)	38 (6.41)	3226 (12.70)	300.03	0.00						
	2	18 (0.95)	387 (1.69)	13 (2.19)	418 (1.65)								
	3	11 (0.58)	437 (1.91)	14 (2.36)	462 (1.82)								
	4	14 (0.74)	435 (1.90)	7 (1.18)	456 (1.80)								
	5	44 (2.32)	833 (3.64)	26 (4.38)	903 (3.56)								
	6	717 (37.82)	18227 (79.58)	475 (80.10)	19419 (76.47)								
	7	17 (0.90)	472 (2.06)	20 (3.37)	509 (2.00)								
Human capital	0	349 (18.41)	1110 (4.85)	24 (4.05)	1483 (5.84)	115.03	0.00						
	1	1133 (59.76)	10612 (46.33)	183 (30.86)	11928 (46.97)								
	2	360 (18.99)	7755 (33.86)	208 (35.08)	8323 (32.78)								
	3	54 (2.85)	3427 (14.96)	178 (30.03)	3659 (14.41)								
Income	1	46 (2.43)	433 (1.89)	12 (2.02)	491 (1.93)	31.37	0.00						
	2	40 (2.11)	393 (1.72)	8 (1.35)	441 (1.74)								
	3	55 (2.90)	846 (3.69)	11 (1.85)	912 (3.59)								
	4	53 (2.80)	1027 (4.48)	14 (2.36)	1094 (4.31)								
	5	1702 (89.77)	20205 (88.22)	548 (92.41)	22455 (88.43)								
Public sector employee	0	1839 (96.99)	20872 (91.13)	513 (86.51)	23224 (91.46)	96.15	0.00						
	1	57 (3.01)	2032 (8.87)	80 (13.49)	2169 (8.54)								
Housing conditions													
Homeowner.	0	104 (5.49)	3878 (16.93)	81 (13.66)	4063 (16.00)	173.17	0.00						
	1	1792 (94.51)	19026 (83.07)	512 (86.34)	21330 (84.00)								
Number of rooms	1	856 (45.15)	5601 (24.45)	175 (29.51)	6632 (26.12)	436.53	0.00						
	2	630 (33.23)	9679 (37.89)	182 (30.69)	9491 (37.38)								
	3	266 (14.03)	6054 (26.43)	168 (28.33)	6488 (25.55)								
	4	114 (6.01)	1932 (8.44)	52 (8.77)	2098 (8.26)								
	5	30 (1.58)	638 (2.79)	16 (2.70)	684 (2.69)								

Internet access	0 1	17772 (93.46) 124 (6.54)	15264 (66.64) 7640 (33.36)	315 (53.12) 278 (46.88)	17351 (68.33) 8042 (31.67)	646.80	0.00						
Access to drinking water	0 1	124 (6.54) 124 (6.54)	15264 (66.64) 7640 (33.36)	315 (53.12) 278 (46.88)	17351 (68.33) 8042 (31.67)	200.03	0.00						
Access to electricity	0 1	324 (17.09) 1572 (82.91)	6538 (28.55) 16366 (71.45)	133 (22.43) 460 (77.57)	8162 (32.14) 17231 (67.86)	120.30	0.00						

Table-4: Likelihood-Ratio and Wald Test for Independent Variables

	Rural						Urban						Total					
	Wald test			Likelihood-ratio test			Wald test			Likelihood-ratio test			Wald test			Likelihood-ratio test		
	Chi^2	Df.	$p>Chi^2$	Chi^2	Df.	$p>Chi^2$	Chi^2	Df.	$p>Chi^2$	Chi^2	Df.	$p>Chi^2$	Chi^2	Df.	$p>Chi^2$	Chi^2	Df.	$p>Chi^2$
Head of household																		
Age																		
61-60 years	7.66	2	0.02	7.95	2	0.02	1.25	2	0.54	1.30	2	0.52	8.78		0.01	9.14		0.01
Over 60 years	27.08	2	0.00	28.63	2	0.00	4.69	2	0.10	4.94	2	0.08	28.42		0.00	30.10		0.00
Marital status																		
Married	23.43	2	0.00	23.63	2	0.00	6.37	2	0.04	6.38	2	0.04	34.99		0.00	35.21		0.00
Gender																		
Man	2.29	2	0.32	2.38	2	0.31	8.12	2	0.02	8.36	2	0.02	10.60		0.01	10.92		0.00
Ethnicity																		
Afro-descendant	24.90	2	0.00	39.84	2	0.00	4.99	2	0.08	5.12	2	0.08	46.92		0.00	68.70		0.00
Black	37.79	2	0.00	70.95	2	0.00	7.50	2	0.02	8.49	2	0.01	67.15		0.00	131.24		0.00
Mulatto	25.79	2	0.00	44.47	2	0.00	1.37	2	0.50	1.47	2	0.48	57.40		0.00	99.83		0.00
Montubio	128.62	2	0.00	182.34	2	0.00	11.06	2	0.00	13.44	2	0.00	157.38		0.00	233.83		0.00
Mestizo	618.78	2	0.00	648.28	2	0.00	14.72	2	0.00	14.13	2	0.00	840.84		0.00	838.18		0.00
White	37.08	2	0.00	56.44	2	0.00	11.27	2	0.00	12.67	2	0.00	60.44		0.00	95.15		0.00
Human capital																		
Basic education	10.70	2	0.01	10.53	2	0.01	0.84	2	0.66	0.79	2	0.67	17.04		0.00	16.64		0.00
Middle education	38.20	2	0.00	38.26	2	0.00	2.76	2	0.25	2.84	2	0.24	59.19		0.00	58.67		0.00
Higher education	17.73	2	0.00	19.72	2	0.00	12.01	2	0.00	14.00	2	0.00	54.53		0.00	62.37		0.00
Income																		
Lower middle income	1.07	2	0.59	1.08	2	0.58	0.27	2	0.88	0.27	2	0.88	0.85		0.66	0.85		0.65
Average income	4.77	2	0.09	4.73	2	0.09	2.55	2	0.28	2.61	2	0.27	6.03		0.05	6.00		0.05
Upper middle income	6.39	2	0.04	6.45	2	0.04	0.53	2	0.77	0.52	2	0.77	5.95		0.05	5.83		0.05
High income	1.98	2	0.37	1.77	2	0.41	0.47	2	0.79	0.52	2	0.77	0.54		0.765	0.52		0.77
Public sector employee																		
Yes	7.91	2	0.02	8.24	2	0.02	1.44	2	0.49	1.36	2	0.51	1.79		0.41	1.84		0.40
Housing conditions																		
Homeowner.																		
Yes	2.83	2	0.24	2.99	2	0.22	18.65	2	0.00	19.96	2	0.00	22.26		0.00	23.91		0.00
Number of rooms.																		
Two	54.26	2	0.00	54.50	2	0.00	26.38	2	0.00	26.28	2	0.00	64.79		0.00	64.78		0.00
Three	51.69	2	0.00	53.93	2	0.00	21.05	2	0.00	21.52	2	0.00	71.11		0.00	74.07		0.00

Four	23.01	2	0.00	24.46	2	0.00	6.55	2	0.04	6.97	2	0.03	24.56		0.00	26.07		0.00
> 5 rooms	12.74	2	0.00	14.65	2	0.00	2.99	2	0.23	3.34	2	0.19	13.53		0.00	15.27		0.00
Intenet access																		
Yes	21.93	2	0.00	24.26	2	0.00	16.93	2	0.00	17.17	2	0.00	94.73		0.00	107.57		0.00
Access to drinking water																		
Yes	197.71	2	0.00	220.79	2	0.00	23.98	2	0.00	22.05	2	0.00	420.70		0.00	455.61		0.00
Access to electricity																		
Yes	19927	2	0.00	213.38	2	0.00	55.37	2	0.00	40.14	2	0.00	267.62		0.00	285.84		0.00
Source: Authors based on INEC (2018).																		

Table-5: Haussmann test of IIA assumptions

	Rural			Urban			Total		
	Chi^2	Df	p>Chi^2	Chi^2	Df	p>Chi^2	Chi^2	Df	p>Chi^2
F&C	9.44	27	0.99	23.15	27	0.68	24.49	27	0.60
LGP	31.64	27	0.25	30.32	27	0.30	25.81	27	0.40
E&O	29.27	27	0.35	15.46	27	0.96	43.73	27	0.02
Source: Authors based on INEC (2018).									

Table 6 presents the results of the multinomial logit model for rural and urban areas and the total population. The "LPG" and "E&O" columns show the coefficients reflecting the probabilistic effect of the explanatory variables on fuel choice. In rural areas, increasing age and married marital status are negatively associated with the use of LPG compared to firewood and charcoal. Conversely, ethnicity, educational level, number of rooms, and access to the Internet, water, and electricity have a positive and significant effect ($p < 0.05$) on LPG use. In urban areas, LPG use is positively associated with belonging to the mestizo ethnic group, having homes with two or three rooms, and access to water and electricity. The use of electricity as a fuel is increasing among the Montubio, mestizo, and white ethnic groups, who have higher education and access to basic services, except for the number of rooms and electricity, which are not significant. In the overall sample, older, married, indigenous, and homeowners were less likely to use LPG compared to CF. On the other hand, higher education and better housing conditions significantly increased the use of LPG and electricity. The likelihood of electricity use increased among household heads with secondary or higher education, non-indigenous individuals, and those with access to Internet and water, while being married reduced this likelihood. These results are consistent with previous studies (Zhang et al., 2023; Waweru et al., 2022; Mothala et al., 2022; Oyedele, 2023; Oyeniran and Isola, 2023; Akeh et al., 2023; Okereke et al., 2023). The main difference is the role of income, which in this analysis is not decisive, although the literature highlights willingness to pay as a relevant factor in the choice of fuels (Yunusa et al., 2024; Nduka, 2021; Azorliade et al., 2022).

Table-6: Multinomial logit (mnl) regression results

Variables	Rural		Urban		Total	
	LGP	E&O	LGP	E&O	LGP	E&O
Head of household						
Age						
31-61 years	-0.37^{**}	-0.38	-0.25	-0.15	-0.34^{**}	-0.26
	(-2.76)	(-1.16)	(-0.99)	(-0.51)	(-2.93)	(-1.39)
> 60	-0.75^{***}	-0.47	-0.57^{*}	-0.42	-0.66^{***}	-0.46^{*}
	(-5.18)	(-1.29)	(-2.04)	(-1.24)	(-5.23)	(-2.20)
Marital status						
Married	-0.35^{***}	-0.57^{**}	-0.11	-0.39	-0.34^{***}	-0.61^{***}
	(-4.64)	(-2.79)	(-0.68)	(-1.95)	(-5.14)	(-5.25)
gender						
Man	-0.05	0.29	0.08	0.43^{*}	-0.08	0.26
	(-0.53)	(1.18)	(0.47)	(2.10)	(-1.08)	(1.95)
Ethnicity						
Afro-Ecuadorian	2.05^{***}	1.55	-0.07	0.96	1.71^{***}	2.22^{***}
	(4.98)	(1.42)	(-0.19)	(1.62)	(6.62)	(5.36)
Black	2.74^{***}	2.94^{***}	1.09^{*}	1.90^{**}	2.62^{***}	3.03^{***}
	(6.12)	(3.91)	(2.09)	(2.74)	(8.05)	(6.73)
Mulato	2.20^{***}	1.47	0.50	0.54	2.15^{***}	1.76^{***}

	(5.04)	(1.32)	(1.17)	(0.76)	(7.54)	(3.36)
Montubio	2.03***	2.45***	1.45*	2.39***	2.08***	2.65***
	(11.20)	(6.16)	(2.56)	(3.33)	(12.26)	(8.61)
Mestizo	1.72***	1.54***	0.76**	1.46***	1.80***	2.01***
	(24.87)	(6.57)	(3.27)	(3.43)	(28.88)	(10.74)
White	2.01***	1.03	0.76	2.04**	1.99***	2.64***
	(6.04)	(0.96)	(1.50)	(3.10)	(7.37)	(6.76)
Human capital						
Basic education	0.30**	-0.03	0.26	0.25	0.36***	0.20
	(3.16)	(-0.10)	(0.91)	(0.58)	(4.10)	(0.83)
Middle education	0.76***	0.82*	0.35	0.75	0.82***	1.11***
	(6.16)	(2.23)	(1.14)	(1.66)	(7.52)	(4.44)
Higher education	0.99***	1.56**	0.67	1.66***	1.13***	2.01***
	(3.91)	(3.25)	(1.90)	(3.39)	(6.26)	(6.83)
Income						
Lower Middle income	0.26	-0.10	-0.31	-0.50	0.20	-0.06
	(0.91)	(-0.15)	(-0.41)	(-0.51)	(0.76)	(-0.11)
Middle income	0.51*	-0.07	-0.30	-1.29	0.40	-0.40
	(2.00)	(-0.11)	(0.43)	(-1.43)	(1.70)	(-0.81)
Upper middle income	0.44	-0.82	0.30	-0.01	0.41	-0.29
	(1.74)	(-1.19)	(-0.43)	(-0.02)	(1.76)	(-0.63)
High income	0.13	-0.42	-0.37	-0.32	0.06	-0.15
	(0.68)	(-0.91)	(-0.68)	(-0.48)	(0.32)	(-0.43)
Public sector employee						
Yes	0.50*	0.91*	-0.31	-0.32	0.20	0.27
	(2.40)	(2.54)	(-1.20)	(-1.07)	(1.24)	(1.27)
Housing conditions						
Homeowner.						
Yes	-0.29	-0.35	0.32	0.86***	-0.38***	0.08
	(-1.67)	(-0.93)	(1.89)	(3.96)	(-3.30)	(0.40)
Number of rooms						
Two	0.55***	0.42	0.46**	-0.12	0.47***	0.001
	(7.36)	(1.87)	(2.74)	(-0.57)	(7.00)	(0.01)
Three	0.71***	0.61*	0.68***	0.24	0.70***	0.34*
	(7.19)	(2.30)	(3.36)	(1.01)	(7.97)	(2.33)
Four	0.63***	0.07	0.50	0.12	0.54***	0.14
	(4.64)	(0.16)	(1.65)	(0.35)	(4.41)	(0.66)
> 5 rooms	0.85***	0.33	0.17	-0.35	0.67**	0.16
	(3.49)	(0.50)	(0.40)	(-0.69)	(3.19)	(0.46)
Internet access						
Yes	0.64***	1.09***	0.32*	0.68***	0.85***	1.27***
	(4.17)	(4.06)	(2.16)	(3.77)	(8.45)	(9.49)
Access to drinking water						
Yes	1.11***	1.26***	0.75***	0.78***	1.34***	1.48***
	(14.00)	(6.37)	(4.88)	(3.77)	(20.42)	(11.80)
Access to electricity						
Yes	1.68***	0.57	3.15***	1.46*	1.91***	0.76*
	(14.09)	(1.46)	(7.33)	(2.09)	(16.32)	(2.31)
Constant	-1.43***	-3.85***	-0.61	-4.07***	-1.37***	-4.71***
	(-4.54)	(-4.88)	(-0.82)	(-3.71)	(-5.19)	(-8.50)
Log-likelihood						
Model	-3983.70		-3029.29		-7192.29	
Intercept-only	-5165.57		-3178.59		-9393.61	
Chi-square						
Deviance(df=4668)	7967.41		6058.58		14384.57	

LR (df=56)	2363.73		298.60		4402.66	
p-value	0.00		0.00		0.00	
R2						
MC Fadden	0.23		0.05		0.23	
MC Fadden (adj.)	0.22		0.03		0.23	
IC						
AIC	8075.41		6166.58		14492.57	
AIC divided by N	0.82		0.41		0.58	
BIC (df=58)	8463.97		6578.46		14931.47	
Observations	9853				25028	
Note: Asterisks ***, **, and * denotes significant levels at the 1%, 5%, and 10% levels, respectively. Dependent variable: type of cooking fuel (F&C= 0 (reference group); LPG = 1; E&O= 2.						

Table 7 presents the marginal effects of the multinomial logit model for rural, urban, and total areas, highlighting the likelihood of LPG use relative to firewood and charcoal (Equation 5). In rural areas, household heads over 60 years of age are 7% less likely to use LPG compared to those under 30, suggesting a lower willingness to abandon traditional fuels. Similarly, married household heads are 3% less likely to use LPG. Regarding ethnicity, all categories show positive and significant effects, indicating that indigenous heads are more likely to continue using firewood and charcoal, possibly for cultural reasons or due to greater availability. Regarding education, those with a secondary level or higher are 8% more likely to use LPG than illiterate individuals. Income is significant at the 10% level. These results are consistent with those of Barik and Padhi (2024) and Chen et al. (2023), who point to per capita income and educational attainment as key factors in the transition to cleaner fuels. Similarly, Gould et al. (2020) highlight those improvements in rural education and income favor the use of clean cooking technologies.

In the rural sector, gender and home ownership do not show statistical significance, while public employment does so at 10%. Although studies such as those by Choudhury et al. (2024) and Okereke et al. (2023) found a significant relationship between gender and fuel type, this association is not confirmed in this case. Regarding housing conditions, it is observed that the probability of using LPG increases with the number of rooms, in contrast to the findings of Akeh et al. (2023), who reported a higher probability of using firewood in larger households in Nigeria. Access to the internet, water, and electricity significantly increases the probability of using LPG by 5%, 10%, and 25%, respectively, with the latter being the most determining factor. These results are consistent with those of Imran and Ozcatalbas (2020), who highlight the role of electricity access in LPG use, and with Belmin et al. (2022), who highlight their contribution to the energy transition and emissions reduction. In contrast, the model shows no significant effects of socioeconomic variables on the use of electricity and other fuels. On the other hand, the use of firewood and charcoal remains more common among indigenous, older, married, privately employed heads of household, those without formal education, and those living in precarious housing conditions.

In urban areas, the probabilities of using firewood, LPG, or electricity as cooking fuels among household heads are reported. Regarding LPG, the number of rooms and access to electricity maintain a positive and significant effect, similar to that observed in rural areas. Households with two to four rooms and access to electricity are more likely to use LPG than firewood and charcoal, with increases of 0.02% and 0.28%, respectively. Structural factors can explain the heterogeneity of these results. Ecuador is a multiethnic country with a high proportion of the population living in energy poverty, which influences fuel choice. It is observed that those over 60 years of age have a positive effect of 0.01% on the use of firewood and charcoal, significant at the 10% level. In contrast, the Montubio ethnic group (-0.02%), three-bedroom homes (-0.01%), and access to water (-0.01%) and electricity (-0.20%) are negatively associated with the use of these fuels ($p < 0.05$). These results reflect a marked difference between urban and rural areas. Although Ishengoma and Igangula (2021) argue that environmental awareness reduces biomass use, in the Ecuadorian case, preferences for traditional fuels may persist due to cultural factors.

The results on the choice of LPG as a fuel in the total population are consistent with the findings of Mothala et al. (2022) for Lesotho. There is a significant negative effect of age (−0.04%), marital status (−1.0%), and home ownership (−3.0%) on LPG use, indicating a lower likelihood of use among men over 60 years of age, who are married and own homes. In contrast, the variables ethnicity, education level, number of rooms, internet access (3%), drinking water (7%), and electricity (19%) show a significant positive effect ($p < 0.05$) on the likelihood of using LPG instead of firewood and charcoal. As in rural areas, indigenous households are less likely to use LPG, while those with higher education are 3% more likely to adopt it. These results are consistent with studies conducted in low- and middle-income countries (Akeh et al., 2023; Oyedele, 2023). Similarly, Paudel et al. (2018) find that electricity availability, higher educational levels, and household wealth favor the transition to cleaner fuels.

Regarding electricity use, men are 1% more likely to use it than women. Household heads with higher education are 3% more likely to use electricity than those without, while homeowners with internet access are only 1% more likely. Conversely, married household heads are 1% less likely to use electricity than single heads, and households with two or three rooms are also 1% less likely. Age and income are not significant, although variables such as ethnicity and access to water and electricity are significant at 10%. Regarding firewood use, positive and significant effects are identified for age, marital status, and homeownership. Household heads over 60 years of age are 3% more likely to use firewood compared to those under 30, and married heads of household who own homes are 2% more likely than single heads who do not own homes. On the contrary, ethnicity, education, number of rooms, and access to basic services negatively impact their use. These results are partially consistent with those of Murshed (2023), Acheampong et al. (2023), and Islam et al. (2023) on the determinants of the adoption of sustainable methods.

Table-7: marginal effects of major cooking fuel choice in Ecuador

Variable	Rural			Urban			Total		
	F&C	LGP	E&O	F&C	LGP	E&O	F&C	LGP	E&O
Head of household									
Age									
31-61 years	0.03^{**}	-0.03^{**}	-0.001	0.003	-0.01	0.002	0.01^{**}	-0.02^{**}	0.001
	(2.97)	(-2.76)	(-0.16)	(1.06)	(-1.02)	(0.54)	(3.15)	(-2.91)	(0.46)
> 60	0.07^{***}	-0.07^{***}	0.003	0.01^{*}	-0.01	0.004	0.03^{***}	-0.04^{***}	0.003
	(5.70)	(-5.59)	(0.54)	(2.22)	(-1.89)	(0.75)	(5.78)	(-5.44)	(1.07)
Marital status									
Married	0.04^{***}	-0.03^{***}	-0.004	0.002	0.01	-0.01^{*}	0.02^{***}	-0.01^{**}	-0.01^{**}
	(4.69)	(-4.01)	(-1.42)	(0.71)	(1.52)	(-2.46)	(5.17)	(-2.83)	(-2.95)
gender									
Man	0.004	-0.01	0.004	-0.001	-0.01^{*}	0.01^{**}	0.004	-0.01^{*}	0.01^{**}
	(0.47)	(-0.92)	(1.54)	(0.12)	(-2.04)	(3.00)	(1.01)	(-2.54)	(3.29)
Ethnicity									
Afro-Ecuadorian	-0.25^{***}	0.25^{***}	-0.002	0.001	-0.03	0.02^{*}	-0.13^{***}	0.12^{***}	0.01

	(-8.92)	(8.46)	(-0.18)	(0.12)	(1-62)	(2.04)	(10.85)	(7.94)	(1.62)
Black	-0.29***	0.28***	0.01	-0.02*	0.001	0.02	-0.16***	0.15***	0.01
	(-14.87)	(11.93)	(0.73)	(-2.53)	(0.09)	(1.66)	(-18.06)	(12.35)	(1.53)
Mulato	-0.26***	0.26	-0.004	-0.01	0.01	0.001	-0.15***	0.15***	-0.004
	(-0.63)	(9.37)	(-0.46)	(-1.27)	(0.86)	(0.09)	(.14,14)	(12.65)	(-0.57)
Montubio	-0.25***	0.24***	0.01	-0.02**	0.0003	0.02*	-0.15***	0.13***	0.02*
	(-17.08)	(14.29)	(0.09)	(-3.17)	(0.02)	(2.03)	(-18.65)	(12.44)	(2.39)
Mestizo	-0.23***	0.23***	0.002	-0.02*	0.001	0.01**	-0.14***	0.13***	0.01*
	(-22.42)	(21.69)	(0.49)	(-2.50)	(0.08)	(2.77)	(-21.94)	(19.02)	(2.15)
White	-0.25***	0.25	-0.01	-0.02	-0.02	0.04**	-0.15***	0.13***	0.02*
	(-10.33)	(10.22)	(-0.85)	(-1.83)	(-1.36)	(2.91)	(-13.24)	(9.22)	(2.28)
Human capital									
Basic education	-0.03**	0.04**	-0.004	-0.005	0.005	-0.0001	-0.02***	0.02***	-0.002
	(-2.96)	(3.11)	(-0.83)	(-0.83)	(0.56)	(1.40)	(-3.76)	(3.49)	(-0.59)
Middle education	-0.08***	0.08***	0.003	-0.01	-0.003	0.01	-0.04***	0.04***	0.01
	(-5.95)	(5.40)	(0.48)	(-1.05)	(-0.37)	(1.40)	(-6.91)	(4.86)	(1.67)
Higher education	-0.10***	0.08***	0.02	-0.01	-0.02*	0.03***	-0.06***	0.03**	0.03***
	(-4.36)	(3.96)	(1.57)	(-1.73)	(-2.12)	(4.00)	(-7.12)	(3.13)	(4.79)
Income									
Lower middle income	-0.03	0.03	-0.01	0.004	0.001	-0.02	-0.01	0.02	-0.01
	(-0.88)	(1.03)	(-0.51)	(0.41)	(0.35)	(1.40)	(-0.74)	(0.92)	(-0.52)
Middle income	-0.05	0.06*	-0.01	0.004	0.01	-0.02	-0.02	0.03*	-0.02
	(-1.91)	(2.13)	(-0.84)	(0.52)	(0.94)	(-1.40)	(-1.61)	(2.32)	(-1.66)
Upper middle income	-0.04	0.06*	-0.02	-0.003	0.01	-0.01	-0.02	0.03*	-0.01
	(-1.62)	(2.15)	(-1.61)	(-0.40)	(0.68)	(-0.56)	(-1.67)	(2.29)	(-1.54)
High income	-0.01	0.02	-0.01	0.005	-0.01	0.001	-0.003	0.01	0.01
	(-0.60)	(0.98)	(-0.99)	(0.80)	(-0.44)	(0.10)	(-0.29)	(0.63)	(-0.61)
Public sector employee									
Yes	-0.05**	0.04*	0.01	0.01	-0.005	-0.0004	-0.01	0.01	0.002
	(-2.74)	(2.17)	(1.31)	(1.06)	(-0.75)	(-0.09)	(-1.31)	(1.01)	(0.53)
Housing conditions									
Homeowner.									
Yes	0.03	-0.02	-0.001	-0.01	-0.01*	0.01***	0.02***	-0.03***	0.01***
	(1.78)	(-1.64)	(-0.27)	(-1.81)	(-1.97)	(4.48)	(3.52)	(-4.84)	(3.77)
Number of rooms									
Two	-0.06***	0.06***	-0.0001	-0.01*	0.02***	-0.02***	-0.03***	0.03***	-0.01***
	(-7.22)	(7.11)	(-0.27)	(-2.57)	(4.87)	(-4.07)	(-6.74)	(7.79)	(-3.81)
Three	-0.08***	0.08***	-0.0001	-0.01**	0.02***	-0.01**	-0.04***	0.04***	-0.01**
	(-7.46)	(7.11)	(-0.02)	(-3.34)	(4.39)	(-3.01)	(-8.19)	(8.33)	(-2.66)
Four	-0.07***	0.07***	-0.01	-0.01	0.02**	-0.01*	-0.03***	0.04***	-0.01*
	(-4.97)	(5.21)	(-1.33)	(-1.83)	(2.77)	(-2.06)	(-4.75)	(5.35)	(-2.37)
> 5 rooms	-0.09***	0.09***	-0.01	-0.003	0.02	-0.02*	-0.03***	0.05***	-0.01*
	(-4.11)	(4.21)	(-0.79)	(-0.39)	(1.76)	(-1.99)	(-3.71)	(4.32)	(-2.10)
Internet access									
Yes	-0.06***	0.05***	0.01	-0.005*	-0.01	0.01***	-0.04***	0.03***	0.01***
	(4.88)	(3.98)	(1.52)	(2.31)	(-1.55)	(3.49)	(-10.22)	(6.24)	(4.74)
Access to drinking water									
Yes	-0.11***	0.10***	0.004	-0.01***	0.01*	0.001	-0.07***	0.07***	0.004*
	(-15.32)	(13.96)	(1.52)	(-4.01)	(2.35)	(0.34)	(-19.76)	(15.81)	(2.02)
Access to electricity									
Yes	-0.23***	0.25***	-0.02	-0.20**	0.28**	-0.08	-0.16***	0.19***	-0.04*
	(-11.29)	(11.02)	(-1.50)	(-3.06)	(3.45)	(-1.38)	(-11.02)	(8.89)	(-2.08)
Note: Asterisks ***, **, and * denotes significant levels at the 1%, 5%, and 10% levels, respectively. Dependent variable: type of cooking fuel (F&C= 0 (reference group); LPG = 1; E&O= 2.									

4.3. Robustness test of results

To assess the robustness of the results, Table 8 presents the estimates from the ordered regression (OL) and generalized ordered logit (GOL) models. Following the literature, an ordinal fuel quality scale was established: 0 for F&C, 1 for LPG, and 2 for E&O. Unlike the OL model, the GOL model allows for greater flexibility by relaxing the proportionality assumption, which makes it possible to capture asymmetric effects of the explanatory variables on the ordinal categories (Williams, 2006). In rural areas, the results of the OL model indicate that increasing age and being married decrease the probability of using cleaner fuels. In contrast, higher educational levels, not belonging to an indigenous ethnic group, being a public employee, having more rooms, and having access to the Internet, water, and electricity increase this probability. In particular, Internet access shows a positive effect, consistent with the findings of Zhang et al. (2023) using a probit model. Similarly, Majumdar et al. (2023) and Akter et al. (2023) highlight that higher social and economic status and access to electricity significantly increase the adoption of cleaner fuels.

In urban areas, household heads of white ethnicity, with higher education, homeowners, and with access to the internet, water, and electricity, are more likely to use clean fuels compared to those of indigenous ethnicity, without formal education, without homeownership, and without access to basic services. In the total population, older age and married marital status are negatively associated with clean fuel use. In contrast, white ethnicity, higher education, a greater number of rooms, and access to essential services are positively associated. Overall, ethnicity, education, and access to basic infrastructure increase the likelihood of electricity use, regardless of geographic setting. These findings are consistent with previous studies such as Zheng (2023), Imran and Ozcatalbas (2020), Silva and Troncoso (2020), and Stoner et al. (2021), which also identify differentiated effects by region and household conditions. In the GOL models, the coefficient $\beta_0/(1,2)$ represents the partial proportional probabilities between F&C and the combination of LPG and electricity. The variables age, marital status, ethnicity, education, and housing conditions in rural areas meet the proportionality assumption, and their results are similar to those of the OL model, indicating a higher probability of using firewood versus clean fuels. In the $\beta_{(0,1)/2}$ model, which compares the use of firewood and LPG versus electricity, no variable is significant at the 5% level, although the signs and magnitudes of the coefficients differ. For example, the coefficient for access to electricity was 1.63 in the OL model, 1.64 in the first GOL model, and -0.43 in the second. The variables ethnicity, education, and number of rooms showed similar patterns. These results are consistent with those reported by Shupler et al. (2021).

In urban areas, the results of the first GOL model indicate that the variables white ethnicity, home ownership, and internet access lose the statistical significance at the 5% level observed in the OL model. However, mestizo ethnicity, number of rooms, and access to water and electricity maintain a positive and significant effect, although the number of rooms was not significant in the OL model. In the second GOL model, the gender coefficient (0.35) is significant, unlike the previous models. Furthermore, white ethnicity, home ownership, and internet access regain significance, with signs consistent with the OL model. In the total sample, the variables age, marital status, and white ethnicity maintain the negative and significant effect of the OL model in the first GOL. It is also noteworthy that home ownership (coefficient -0.35) gains significance, indicating that those who meet all the housing conditions tend to use more firewood than gas or electricity. In the second GOL, age loses significance, while gender becomes significant. Variables such as marital status (-0.29), white ethnicity (0.94), higher education (0.98), and access to the internet (0.44) and water (0.23) maintain their effect—however, home ownership and the number of rooms change sign. Income is not significant in any model, suggesting that other variables better explain fuel choice. Several studies have documented that the use of more polluting fuels is associated with greater health and environmental risks (Fadly et al., 2023; Yokoo et al., 2023; Sreeja et al., 2023; Liao et al., 2023).

Table-8: Robust checks

Variable	*Rural*			*Urban*			*Total*		
	ORL	*GOLR*		*ORL*	*GOLR*		*ORL*	*GOLR*	
		$\beta_{0/1,2}$	$\beta_{0,1/2}$		$\beta_{0/1,2}$	$\beta_{0,1/2}$		$\beta_{0/1,2}$	$\beta_{0,1/2}$
Head of household									
Age									
31-61 years	-0.32***	-0.37**	-0.06	-0.02	-0.25	0.09	-0.20*	-0.34**	0.06
	(-2.69)	(-2.81)	(-0.18)	(-0.16)	(-0.99)	(0.52)	(-2.33)	(-2.94)	(0.41)
> 60	-0.63***	-0.75***	0.19	-0.11	-0.57*	0.15	-0.43***	-0.67***	0.17
	(-4.84)	(-5.22)	(0.56)	(-0.70)	(-2.06)	(0.74)	(-4.42)	(-5.27)	(1.01)
Marital status									
Married	-0.33***	-0.34***	-0.28	-0.22*	-0.11	-0.28*	-0.33***	-0.34***	-0.29**
	(-4.84)	(-4.63)	(-1.46)	(-2.36)	(-0.69)	(-2.43)	(-6.06)	(-5.15)	(-2.98)
Gender									
Man	0.01	-0.04	0.35	0.25*	0.08	0.35**	0.06	-0.08	0.34**
	(0.12)	(-0.47)	(1.48)	(2.54)	(0.50)	(2.83)	(1.01)	(-1.02)	(3.10)
Ethnicity									
Afro-Ecuadorian	1.89***	2.05***	-0.09	0.52	-0.05	1.04*	1.85***	1.73***	0.81*
	(5.36)	(4.99)	(-0.08)	(1.53)	(-0.14)	(2.23)	(8.86)	(6.70)	(2.41)
Black	2.41***	2.67***	0.69	0.95***	1.11*	0.83	2.26***	2.58***	0.74*
	(7.30)	(6.01)	(1.10)	(2.95)	(2.14)	(1.75)	(11.55)	(7.98)	(2.28)
Mulato	1.91***	2.20***	-0.32	0.29	0.51	0.03	1.82***	2.14***	-0.06
	(5.66)	(5.03)	(-0.31)	(0.86)	(1.18)	(0.06)	(9.13)	(7.53)	(-0.14)
Montubio	2.04***	2.01***	0.85*	1.08***	1.32*	0.94*	2.09***	2.04***	0.89***
	(11.95)	(11.16)	82.32)	(3.60)	(2.38)	(2.09)	(14.35)	(12.13)	(3.34)
Mestizo	1.65***	1.72***	0.19	0.75***	0.77***	0.71*	1.77***	1.81***	0.50**
	(24.61)	(24.87)	(0.80)	(3.86)	(3.31)	(1.97)	(29.92)	(29.01)	(2.70)
White	1.80***	2.00***	-0.58	1.23***	0.81	1.29***	2.12***	2.03***	0.94**
	(6.34)	(6.03)	(-0.56)	(4.10)	(1.60)	(2.99)	(11.11)	(7.54)	(3.23)
Human capital									
Basic education	0.29***	0.29**	-0.23	0.22	0.27	0.02	0.40***	0.35***	-0.09
	(3.09)	(3.00)	(-0.73)	(0.93)	(0.97)	(0.05)	(4.71)	(4.04)	(-0.39)
Middle education	0.71***	0.76***	0.21	0.51*	0.37	0.43	0.84***	0.83***	0.39
	(6.14)	(6.16)	(0.60)	(2.09)	(1.23)	(1.25)	(8.64)	(7.62)	(1.70)
Higher education	0.97***	1.00***	0.74	1.05***	0.72*	1.01**	1.23***	1.16***	0.98***
	(4.89)	(3.97)	(1.75)	(4.13)	(2.04)	(2.93)	(10.36)	(6.45)	(4.07)
Income									
Lower middle income	0.18	0.26	-0.35	-0.22	-0.36	-0.16	0.12	0.21	-0.23
	(0.66)	(0.93)	(-0.53)	(-0.46)	(-0.48)	(-0.24)	(0.49)	(0.79)	(-0.49)
Middle income	0.35	0.39	-0.53	-0.45	-0.35	-0.99	0.13	0.38	-0.76
	(1.44)	(1.93)	(-0.91)	(-1.16)	(-0.54)	(-1.51)	(0.66)	(1.64)	(-1.76)
Upper middle income	0.22	0.43	-1.20	-0.08	0.27	-0.31	0.15	0.41	-0.66
	(0.93)	(1.71)	(-1.83)	(-0.20)	(0.39)	(-0.59)	(0.74)	(1.78)	(-1.64)
High income	0.03	0.13	-0.54	-0.11	-0.40	0.05	-0.01	0.07	-0.19
	(0.14)	(0.67)	(-1.26)	(-0.33)	(-0.74)	(0.11)	(-0.07)	(0.37)	(-0.65)
Public sector employee									
Yes	0.44**	0.51*	0.47	-0.10	-0.32	-0.01	0.10	0.20	0.08
	(2.75)	(2.45)	(1.56)	(-0.74)	(-1.21)	(-0.07)	(1.02)	(1.20)	(0.58)
Housing conditions									
Homeowner.									
Yes	-0.20	-0.26	-0.08	0.44***	0.34*	0.55***	-0.01	-0.35**	0.43***
	(-1.34)	(-1.46)	(-0.23)	(4.21)	(2.00)	(3.86)	(-0.11)	(-3.08)	(3.32)
Number of rooms									
Two	0.51***	0.55***	-0.05	-0.17	0.45**	-0.57***	0.25***	0.46***	-0.44***
	(7.07)	(7.34)	(-0.22)	(-1.59)	(2.68)	(-4.31)	(4.24)	(6.80)	(-3.97)
Three	0.64***	0.71***	0.01	-0.04	0.67***	-0.42**	0.42***	0.69***	-0.32**
	(7.03)	(7.22)	(0.03)	(-0.33)	(3.31)	(-3.11)	(5.96)	(7.88)	(-2.68)

Four	0.52***	0.62***	-0.45	-0.04	0.49	-0.36	0.32***	0.53***	-0.37*
	(4.18)	(4.56)	(-1.13)	(-0.26)	(1.62)	(-1.93)	(3.31)	(4.33)	(-2.18)
> 5 rooms	0.67**	0.82***	-0.39	-0.25	0.15	-0.51	0.37*	0.63**	-0.47
	(3.22)	(3.33)	(-0.63)	(-0.95)	(0.35)	(-1.67)	(2.36)	(2.99)	(-1.74)
Internet access									
Yes	0.49***	0.65***	0.47*	0.34***	0.33*	0.36***	0.56***	0.86***	0.44***
	(4.20)	(4.19)	(2.09)	(4.04)	(2.23)	(3.49)	(8.97)	(8.49)	(4.83)
Access to drinking water									
Yes	0.96***	1.12***	0.26	0.40***	0.74***	0.05	1.14***	1.34***	0.23*
	(13.41)	(14.06)	(1.39)	(3.48)	(4.87)	(0.35)	(19.66)	820.47)	(2.06)
Access to electricity									
Yes	1.63***	1.64***	-0.43	2.91***	3.05***	-1.40*	1.92***	1.86***	-0.48
	(13.98)	(13.95)	(-1.10)	(6.43)	(7.23)	(-2.23)	(16.96)	(16.18)	(-1.47)
Constant	1.34***	-1.39***	-3.75***	0.45	-0.48	-3.83***	1.69***	-1.33***	-4.49***
	(4.63)	(-4.46)	(-5.14)	(0.74)	(-0.65)	(-4.42)	(7.44)	(-5.07)	(-0.19)
Chi-square									
Deviance (df=4668)	8126.73	7969.57		6154.28	6058.34		14789.9	14397.73	
LR (df=56)	2204.40	2361.56		202.90	298.84		3997.26	4399.49	
p-value	0.00	0.00		0.00	0.00		0.00	0.00	
R2									
MCFadden	0.21	0.23		0.03	0.04		0.21	0.23	
MCFadden (adjusted)	0.21	0.22		0.02	0.03		0.21	0.23	
IC									
AIC	8182.73	8077.57		6210.75	6166.34		14845.97	14495.73	
AIC divided by N	0.83	0.82		0.41	0.41		0.59	0.58	
BIC (df=58)	8284.21	8466.13		6423.84	6578.22		15073.55	14934.63	
Observation	9853	9853		15175	15175		25028	25028	

Note: Asterisks ***, **, and * denotes significant levels at the 1%, 5%, and 10% levels, respectively.
Dependent variable: type of cooking fuel (F&C= 0 (reference group); LPG = 1; E&O= 2.

The analysis of the results reveals that, for the total population, the effect of the explanatory variables in the GOL models is asymmetric and varies across models, especially for age, sex, home ownership, number of rooms, and access to electricity. This divergence is consistent given that the comparison categories differ across models, so the variables are not expected to have identical effects. As Williams (2016) points out, a lower probability of belonging to one category does not necessarily imply an equivalent effect on the probability of belonging to another. Therefore, estimating only the OL model would have limited the ability to capture the variability in the effects. In terms of fit, the model for the total population shows the best performance according to the maximum likelihood criterion. However, it presents a greater deviation from the models for urban and rural areas. The likelihood ratio test confirms a significant improvement in all GOL models compared to the OL models. However, when considering the AIC and BIC criteria for public policy decision-making, the model corresponding to the urban area is the most appropriate.

5. Conclusions and policy implications

This study examined the effect of socioeconomic factors on cooking fuel choices in Ecuadorian households, differentiating between urban and rural areas. Three fuel types were analyzed: firewood and charcoal (F&C), LPG, and electricity, based on individual and household data, using multinomial logit and generalized ordered logit models. The results show that the transition to cleaner fuels is determined by a combination of demographic, economic, cultural, and infrastructure factors. In rural areas, the education of the head of household, the number of rooms, and access to basic services (water, electricity, and internet) are the main drivers of LPG use. In contrast, households headed by older adults, indigenous people, and those without education or access to services are more likely to use firewood and charcoal. In urban areas, LPG use is positively associated with better housing conditions and access to services. The use of electricity as a fuel is more common among men, white heads of household, those with higher education, and those with small homes and internet access. However, access to electricity is not significant, highlighting structural barriers to the adoption of cleaner technologies.

At the national level, positive determinants of LPG use include education, income, housing conditions, and access to services, while older age, indigenous ethnicity, and married marital status have negative effects. The likelihood of electricity use increases in households with better social and economic conditions, demonstrating that such improvements favor the use of more sustainable fuels. These findings are consistent with studies conducted in other contexts (Ishengoma and Igangula, 2021; Waweru et al., 2022; Mothala et al., 2022; Chen et al., 2023; Oyedele, 2023; Oyeniran and Isola, 2023), although they differ in the impact of gender and marital status, which in the Ecuadorian case are negatively associated with electricity use. Furthermore, GOL models reveal asymmetric effects in proportional probabilities, highlighting the complexity of the energy decision-making process in households.

The policy implications derived from this research suggest the need to strengthen access to basic services in rural areas, such as water, electricity, and internet, to facilitate the transition to cleaner fuels. Furthermore, it is recommended that induction cooktop promotion policies be redesigned through economic incentives, partnerships with manufacturers, technical training, and awareness-raising campaigns tailored to the socioeconomic context of households. It is essential to implement differentiated strategies between urban and rural areas, given their marked structural differences. Furthermore, it proposes a review of the current LPG subsidy scheme, which, while serving a social function, generates considerable fiscal and environmental costs. Finally, it encourages research that analyzes the comparative profitability of induction cooktops versus LPG cooktops and delves deeper into the external factors that influence the use of sustainable fuels in developing countries.

References:

Acheampong, A. O., Opoku, E. E. O., & Dogah, K. E. (2023). The political economy of energy transition: The role of globalization and governance in the adoption of clean cooking fuels and technologies. *Technological Forecasting and Social Change*, 186, 122156.

Akeh, G. I., Adamu, B. M., Adamu, H., & Ade, S. M. (2023). Determinants of Household Cooking Energy Choice in Public Housing Estates in Maiduguri, Borno State, North-East Nigeria. *Journal of Energy Research and Reviews*, 13(3), 61-75.

Akter, S., Mathew, N. M., & Fila, M. E. (2023). The impact of an improvement in the quality and reliability of rural residential electricity supply on clean cooking fuel adoption: Evidence from six energy poor Indian states. *World Development*, *172*, 106366.

Alem, Y., Beyene, A. D., Köhlin, G., & Mekonnen, A. (2016). Modeling household cooking fuel choice: A panel multinomial logit approach. *Energy Economics*, *59*, 129-137.

Ali, E., Yaotse, K., Obeng, E. O. B., Gyamfi, S., Osman, M. S., Adoko, T., & Narra, S. (2024). Determinants of household cooking fuel choices: Does proximity to mine site matter?. *Energy for Sustainable Development*, *82*, 101545.

Ali, J., & Khan, W. (2022). Factors affecting access to clean cooking fuel among rural households in India during COVID-19 pandemic. *Energy for Sustainable Development*, 67, 102-111.

Amadu, I., Seidu, A.-A., Mohammed, A., Duku, E., Miyittah, M. K., Ameyaw, E. K., Hagan, J. E., Musah, M. H., & Ahinkorah, B. O. (2023). Assessing the combined effect of household cooking fuel and urbanicity on acute respiratory symptoms among under-five years in sub-Saharan Africa. *Heliyon*, 9(6), e16546.

Aminu, R. O., Dzudzor, M. A., Oyawole, F. P., & Afolayan, S. O. (2024). Understanding the determinants of household cooking-fuel choice in sub-Saharan Africa: evidence from Nigeria. *Clean Energy*, *8*(2), 144-155.

Arowolo, A. O., Ibrahim, S. B., Sanusi, M. M., Ayinde, A. F. O., & Sheleru, K. M. (2018). Households 'cooking energy preferences and determinants of transition to non-biomass fuels in rural Nigeria. *Journal of Sustainable Development in Africa*, *20*(1).

Azorliade, D. A., Twerefou, D. K., & Dovie, D. B. K. (2022). The Impact of Household Cooking Fuel Choice on Healthcare Expenditure in Ghana. *Frontiers in Environmental Science*, *10*, 861204.

Bai, C., Sun, Z., Feng, C., & Xiao, W. (2024). Human capital and cooking fuel choices in rural China: Perspective from cognitive and noncognitive skills. *Energy Policy*, *184*, 113896.

Barik, N., & Padhi, P. (2024). The determinants of household cooking fuel transition: evidence from rural India. *Environment, Development and Sustainability*, 1-29.

Barua, S., Aziz, S., Arefin, S., & Chowdhury, S. A. (2025). The moderating effects of land and cattle ownership on the use of biomass as cooking fuel in Bangladesh. *Biomass and Bioenergy*, 197, 107803.

Bawakyillenuo, S., Agbelie, I. S., Crentsil, A. O., & Danquah, S. K. (2025). Preferences of future cooking fuel types among urban and peri-urban households in Greater Accra Region of Ghana: Business-as-usual or sustainable pathways? *Energy Policy*, 199, 114525.

Belmin, C., Hoffmann, R., Pichler, P.-P., & Weisz, H. (2022). Fertility transition powered by women's access to electricity and modern cooking fuels. *Nature Sustainability*, 5(3), Article 3.

Cabiyo, B., Ray, I., & Levine, D. I. (2020). The refill gap: Clean cooking fuel adoption in rural India. *Environmental Research Letters*, 16(1), 014035.

Chen, Y., Li, S., Zhou, T., Lei, X., Liu, X., & Wang, Y. (2023). Household cooking energy transition in rural mountainous areas of China: Characteristics, drivers, and effects. *Journal of Cleaner Production*, 385, 135728.

Choudhury, D. K., Guptha, S. K., & Gurung, R. (2024). Determinants of household choice of cooking fuel in India: evidence from National Family Health Survey 5. *International Journal of Social Economics*. *In Press*.

Danlami, A. H., Applanaidu, S. D., & Islam, R. (2018). An analysis of household cooking fuel choice: A case of Bauchi State, Nigeria. *International Journal of Energy Sector Management*, 12(2), 265-283.

Davi-Arderius, D., Obaco, M., & Alvarado, R. (2023). Household Socioeconomic Determinants of Clean Cooking Program in Ecuador. *Energy Sources, Part B: Economics, Planning, and Policy*, 18(1), 2160525.

Dongzagla, A., & Adams, A.-M. (2022). Determinants of urban household choice of cooking fuel in Ghana: Do socioeconomic and demographic factors matter? *Energy*, 256, 124613.

Fadly, D., Fontes, F., & Maertens, M. (2023). Fuel for food: Access to clean cooking fuel and food security in India. *Food Security*, *15*(2), 301-321.

Faizan, M. A., & Thakur, R. (2019). Association Between Solid Cooking Fuels and Respiratory Disease Across Socio-Demographic Groups in India. *Journal of Health & Pollution*, 9(23), 190911.

Farsi, M., Filippini, M., & Pachauri, S. (2007). Fuel choices in urban Indian households. *Environment and Development Economics*, *12*(6), 757-774.

Gill-Wiehl, A., Ray, I., & Kammen, D. (2021). Is clean cooking affordable? A review. *Renewable and Sustainable Energy Reviews*, 151, 111537.

Gould, C. F., Bejarano, M. L., Kioumourtzoglou, M.-A., Lee, A. G., Pillarisetti, A., Schlesinger, S. B., Terán, E., Valarezo, A., & Jack, D. W. (2023). Widespread Clean Cooking Fuel Scale-Up and under-5 Lower Respiratory Infection Mortality: An Ecological Analysis in Ecuador, 1990-2019. *Environmental Health Perspectives*, 131(3), 37017.

Gould, C. F., Hou, X., Richmond, J., Sharma, A., & Urpelainen, J. (2020). Jointly modeling the adoption and use of clean cooking fuels in rural India. *Environmental Research Communications*, 2(8), 085004.

Gould, C. F., Schlesinger, S. B., Molina, E., Bejarano, M. L., Valarezo, A., & Jack, D. W. (2020). Household fuel mixes in peri-urban and rural Ecuador: Explaining the context of LPG, patterns of continued firewood use, and the challenges of induction cooking. *Energy Policy,* 136, 111053.

Gould, C. F., Schlesinger, S. B., Molina, E., Bejarano, M. L., Valarezo, A., & Jack, D. W. (2020). Household fuel mixes in peri-urban and rural Ecuador: Explaining the context of LPG, patterns of continued firewood use, and the challenges of induction cooking. *Energy policy*, *136*, 111053.

Gould, C. F., Schlesinger, S., Toasa, A. O., Thurber, M., Waters, W. F., Graham, J. P., & Jack, D. W. (2018). Government policy, clean fuel access, and persistent fuel stacking in Ecuador. *Energy for Sustainable Development*, 46, 111-122.

Greene, W. H. (2012). Econometric Analysis 7th Edition International edition. Essex, England: Pearson Education Limited.

Gu, J. (2022). Importance of neighbors in rural households' conversion to cleaner cooking fuels: The impact and mechanisms of peer effects. *Journal of Cleaner Production*, 379, 134776.

Gunasegaran, M. K., & Hasanuzzaman, M. (2025). Assessing the Transition in Commercial Cooking Fuels: An Energy, Economic, and Environmental Analysis of LPG and Electricity in Malaysia. *Energy Nexus*, 100428.

Gyamfi, K. S., Gaura, E., Brusey, J., Trindade, A. B., & Verba, N. (2020). Understanding household fuel choice behaviour in the Amazonas State, Brazil: effects of validation and feature selection. *Energies*, *13*(15), 3857.

Hosier, R. H., & Dowd, J. (1987). Household fuel choice in Zimbabwe: an empirical test of the energy ladder hypothesis. *Resources and energy*, *9*(4), 347-361.

Hsu, E., Forougi, N., Gan, M., Muchiri, E., Pope, D., & Puzzolo, E. (2021). Microfinance for clean cooking: What lessons can be learned for scaling up LPG adoption in Kenya through managed loans? *Energy Policy*, 154, 112263.

Huang, W., Li, S., Yang, H., & Yang, H. (2024). Does family care promote clean cooking energy choices for older persons? –Analysis in light of home-based care in rural China. *Energy for Sustainable Development*, *79*, 101402.

Imran, M., & Ozcatalbas, O. (2020). Determinants of household cooking fuels and their impact on women's health in rural Pakistan. *Environmental Science and Pollution Research International*, 27(19), 23849-23861.

Ishengoma, E. K., & Igangula, N. H. (2021). Determinants of household choice of cooking energy-mix in a peri-urban setting in Tanzania. *Energy for Sustainable Development*, 65, 25-35.

Islam, S., Rana, J., & Shupler, M. (2023). Deepened socioeconomic inequality in clean cooking fuel use in India from 2005 to 2006 to 2015–2016. *Heliyon*, e17041.

Kapsalyamova, Z., Mishra, R., Kerimray, A., Karymshakov, K., & Azhgaliyeva, D. (2021). Why energy access is not enough for choosing clean cooking fuels? Evidence from the multinomial logit model. *Journal of Environmental Management*, *290*, 112539.

Kumar, R. (2024). Validez de la hipótesis de la escalera energética según el tipo de residencia: Un análisis probit ordenado para hogares indios. *The Indian Economic Journal*, 00194662241265482.

Li, H., Mu, W., Chen, T., & Wu, J. (2024). A social network perspective on household cooking fuel transition: Evidence from China. *Energy Economics*, *131*, 107314.

Li, M., & Zhou, S. (2023). Pollutive cooking fuels and rural labor supply: Evidence from a large-scale population census in China. *Energy Policy*, 183, 113780.

Li, X., Duan, C., Chen, Q., Xiao, J., & (Jim) Zhang, J. (2023). Associations between cooking fuels and hypertension prevalence in Chinese adults: A prospective cohort analysis focusing on fuel transitioning. *Environment International*, 175, 107953.

Liao, H., Chen, T., Tang, X., & Wu, J. (2019). Fuel choices for cooking in China: Analysis based on multinomial logit model. *Journal of Cleaner Production*, *225*, 104-111.

Liao, W., Liu, X., Kang, N., Song, Y., Wang, L., Yuchi, Y., Huo, W., Mao, Z., Hou, J., & Wang, C. (2023). Associations of cooking fuel types and daily cooking duration with sleep quality in rural adults: Effect modification of kitchen ventilation. *Science of The Total Environment*, 854, 158827.

Liu, Y., Chen, J., Zhao, L., & Liao, H. (2023). Rural photovoltaic projects substantially prompt household energy transition: Evidence from China. *Energy*, 275, 127505.

Liu, Z., Wang, M., Xiong, Q., & Liu, C. (2020). Does centralized residence promote the use of cleaner cooking fuels? Evidence from rural China. *Energy Economics*, 91, 104895.

Long, J. S., & Freese, J. (2006). *Regression models for categorical dependent variables using Stata* (Vol. 7). Stata press.

Ma, C., & Liao, H. (2018). Income elasticity of cooking fuel substitution in rural China: Evidence from population census data. *Journal of Cleaner Production*, 199, 1083-1091.

Majumdar, D., Koley, M., & Chatterjee, P. (2023). Socio-economic factors and clean cooking fuel consumption in India: A household level study. *Energy for Sustainable Development*, *76*, 101298. https://doi.org/10.1016/j.esd.2023.101298

Masera, O. R., Saatkamp, B. D., & Kammen, D. M. (2000). From linear fuel switching to multiple cooking strategies: a critique and alternative to the energy ladder model. *World development*, *28*(12), 2083-2103.

Mbaka, C. K., Gikonyo, J., & Kisaka, O. M. (2019). Households' energy preference and consumption intensity in Kenya. *Energy, Sustainability and Society*, 9(1), 20.

McLean, E. V., Bagchi-Sen, S., Atkinson, J. D., & Schindel, A. (2019). Household dependence on solid cooking fuels in Peru: An analysis of environmental and socioeconomic conditions. *Global Environmental Change*, 58, 101961.

Mothala, M., Thamae, R., & Mpholo, M. (2022). Determinants of household energy fuel choice in Lesotho. *Journal of Energy in Southern Africa*, *33*(2), 24-34.

Murshed, M. (2023). The relevance of reducing income inequality for eliminating urban-rural divide in clean cooking fuel accessibility: Evidence from Latin America and the Caribbean. *Energy*, 278, 127718.

Nduka, E. (2021). How to get rural households out of energy poverty in Nigeria: A contingent valuation. *Energy Policy*, 149, 112072.

Nuño, N., Mäusezahl, D., Hartinger, S. M., Riley-Powell, A. R., Verastegui, H., Wolf, J., Muela, J., & Paz-Soldán, V. A. (2023). Acceptance and uptake of improved biomass cookstoves in Peru – Learning from system level approaches to transform large-scale cooking interventions. *Energy Research & Social Science*, 97, 102958.

Okereke, C., Onyeneke, R. U., Ijeoma, S., Fadero, T., Ahanotu, K., & Anieze, E. E. (2023). Attitude, knowledge and perception of choice of cooking fuels: Evidence from two large communities in South-east Nigeria. *Environmental Progress & Sustainable Energy*, 42(1), e13983.

Onyeneke, R. U., Chidiebere-Mark, N. M., Ankrah, D. A., & Onyeneke, L. U. (2023). Determinants of access to clean fuels and technologies for cooking in Africa: A panel autoregressive distributed lag approach. *Environmental Progress & Sustainable Energy*, 42(3), e14147.

Oyedele, O. (2023). Determinants of Household Cooking Energy Choice: Are Such Choices Influenced by Health Outcomes? *International Journal of Energy Economics and Policy*, 13(2).

Oyeniran, I. W., & Isola, W. A. (2023). Patterns and determinants of household cooking fuel choice in Nigeria. *Energy*, 127753.

Pan, M., Huang, L., Song, D., & Qian, X. (2023). Can intellectual property rights protection reduce air pollution? A quasi-natural experiment from China. *Structural Change and Economic Dynamics*, 65, 210-222.

Pantelic, J., Son, Y. J., Staven, B., & Liu, Q. (2023). Cooking emission control with IoT sensors and connected air quality interventions for smart and healthy homes: Evaluation of effectiveness and energy consumption. *Energy and Buildings*, 286, 112932.

Paudel, U., Khatri, U., & Pant, K. P. (2018). Understanding the determinants of household cooking fuel choice in Afghanistan: A multinomial logit estimation. *Energy,* 156, 55-62.

Pelz, S., Chindarkar, N., & Urpelainen, J. (2021). Energy access for marginalized communities: Evidence from rural North India, 2015–2018. *World Development*, 137, 105204.

Perros, T., Allison, A. L., Tomei, J., & Parikh, P. (2022). Behavioural factors that drive stacking with traditional cooking fuels using the COM-B model. *Nature Energy*, *7*(9), 886-898.

Pu, F., Li, C., Zhang, X., Cao, X., Yang, Z., Hu, Y., Xu, X., Ma, Y., Hu, K., & Liu, Z. (2023). Transition of cooking fuel types and mortality risk in China, 1991–2015. *Science of The Total Environment*, 869, 161654.

Qiu, X., Jin, J., He, R., & Zhang, C. (2023). Do solid fuels for cooking lead to an increased prevalence of respiratory disease? Empirical evidence from rural China. *Energy for Sustainable Development*, 74, 297-308.

Severo, E. A., De Guimarães, J. C. F., Wanderley, L. S. O., Gueiros, M. M. B., & Jabbour, C. J. C. (2023). Influence of the COVID-19 pandemic on the use of social media on awareness' socio-environmental and sustainable consumption: Consolidating lessons from the pandemic. *Environmental Development*, 46, 100865.

Sharma, K., & Rahman, N. (2025). Do cooking characteristics and household amenities matter for cooking fuel choice, clean fuel transition, and clean fuel consumption in India? *Energy Policy*, *198*, 114510.

Shupler, M., Mangeni, J., Tawiah, T., Sang, E., Baame, M., Anderson de Cuevas, R., ... & Pope, D. (2021). Modelling of supply and demand-side determinants of liquefied petroleum gas consumption in peri-urban Cameroon, Ghana and Kenya. *Nature Energy*, *6*(12), 1198-1210.

Silva, A. S. da, & Troncoso, K. S. (2020). Public policies to promote the elimination of polluting fuels for cooking in households: Lessons learned from Bolivia, Ecuador and El Salvador. ISEE Conference Abstracts.

Sreeja, A., Dhengle, A., Kumar, D., & Pradhan, A. (2023). Does access to clean cooking fuels reduce environmental degradation? Evidence from BRICS nations. *Environmental Science and Pollution Research*, 1-11.

Stoner, O., Lewis, J., Martínez, I. L., Gumy, S., Economou, T., & Adair-Rohani, H. (2021). Household cooking fuel estimates at global and country level for 1990 to 2030. *Nature Communications*, *12*(1), 5793.

Valarezo, A., Dávila, L., Bejarano, M. L., Nolivos, I., Molina, E., Schlesinger, S. B., Gould, C. F., & Jack, D. W. (2023). Resilient clean cooking: Maintaining household clean cooking in Ecuador during the COVID-19 pandemic. *Energy for Sustainable Development*, 74, 349-360.

Van der Kroon, B., Brouwer, R., & Van Beukering, P. J. (2013). The energy ladder: Theoretical myth or empirical truth? Results from a meta-analysis. *Renewable and sustainable energy reviews*, *20*, 504-513.

Wang, X., Fielding, K. S., & Dean, A. J. (2023). Psychological ownership of nature: Relationships with pro-environmental intentions in less environmentally-oriented individuals. *Personality and Individual Differences,* 213, 112304.

Wang, Y., Liang, L., Xu, W., Liu, C., Cheng, H., Liu, Y., Zhang, G., Xu, X., Yu, D., Wang, P., Song, Q., Liu, J., & Cheng, Y. (2024). Influence of meteorological factors on open biomass burning at a background site in Northeast China. *Journal of Environmental Sciences*, 138, 1-9.

Wassie, Y. T., Rannestad, M. M., & Adaramola, M. S. (2021). Determinants of household energy choices in rural sub-Saharan Africa: An example from southern Ethiopia. *Energy*, 221, 119785.

Waweru, D., Mose, N., & Otieno, S. (2022). Household Energy Choice in Kenya: An Empirical Analysis of the Energy Ladder Hypothesis. *Journal of Energy Research and Reviews*, 12-19.

Williams, R. (2006). Generalized ordered logit/partial proportional odds models for ordinal dependent variables. *Stata Journal*, 6, 58–82.

Williams, R. (2016). Understanding and interpreting generalized ordered logit models. The *Journal of Mathematical Sociology*, 40(1), 7-20.

Xu, T., Ye, X., Lu, X., Lan, G., Xie, M., Huang, Z., Wang, T., Wu, J., Zhan, Z., & Xie, X. (2023). Association between solid cooking fuel and cognitive decline: Three nationwide cohort studies in middle-aged and older population. *Environment International*, 173, 107803.

Yao, M., & Liu, L. (2025). Does population aging aggravate energy poverty: An examination of clean cooking fuel choice in rural China. *Economic Analysis and Policy,* 86, 210-230.

Yokoo, H.-F., Arimura, T. H., Chattopadhyay, M., & Katayama, H. (2023). Subjective risk belief function in the field: Evidence from cooking fuel choices and health in India. *Journal of Development Economics*, *161*, 103000.

Yunusa, S. U., Mensah, E., Preko, K., Narra, S., Saleh, A., & Sanfo, S. (2024). Assessing the nexus between household dynamics and cooking energy choice: evidence from Kaduna state, northwestern Nigeria. *Energy Nexus*, 100310.

Zhang, H., Yang, F., Chandio, A. A., Liu, J., Twumasi, M. A., & Ozturk, I. (2023). Assessing the effects of internet technology use on rural households' cooking energy consumption: Evidence from China. *Energy*, *284*, 128726.

Zhang, X. B., & Hassen, S. (2017). Household fuel choice in urban China: evidence from panel data. *Environment and Development Economics*, *22*(4), 392-413.

Zheng, L. (2023). Impact of off-farm employment on cooking fuel choices: Implications for rural-urban transformation in advancing sustainable energy transformation. *Energy Economics*, *118*, 106497.

Zhou, B., & Ding, H. (2023). How public attention drives corporate environmental protection: Effects and channels. *Technological Forecasting and Social Change*, 191, 122486.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

The authors have no conflict of interest.